An Efficient, Globally Optimal Two-Step Seamline Detection Method for Batch Satellite Orthorectified Images

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***Abstract*—Conventional pixel-level seamline detection algorithms exhibit exponential time complexity on large, batch-mode remote-sensing mosaics, making it difficult to achieve an optimal trade-off between accuracy and efficiency. This article introduces a globally optimal and highly efficient seamline detection framework. First, a preliminary seamline network is generated by iteratively clipping valid orthoimage regions with a Voronoi diagram, and image blocks are extracted only within overlap areas to markedly reduce data volume. Second, a cost graph constructed on down-sampled blocks is traversed in a reverse-diagonal Z-pattern; a "local entropy-gradient" composite cost function is applied, and a linear-time dynamic-programming (DP) scheme rapidly produces coarse seamlines that bypass texture-rich regions and confine the search space to a narrow band. Third, a buffer centered on the coarse seamline is created, within which an enhanced Dijkstra algorithm performs pixel-level refinement to accurately avoid complex obstacles. Experiments on the GF-7 data set demonstrate that, compared with five representative methods—SMP-DP, A*, Dijkstra, graph-cut, and OrthoVista—the proposed approach improves geometric accuracy by 14.46%, 58.69%, 50.20%, 17.79%, and 69.30%, respectively; processing efficiency is increased by 12.74%, 19.19%, 49.89%, >500%, and 83.72%, respectively. The algorithm has successfully mosaicked 627 GF-7 scenes covering the entire Henan Province, and has yielded similarly favorable results on ZY-3, GF-1, and GF-3 imagery, underscoring its high applicability and robustness for multisource, large-format remote-sensing production.**

***Index Terms*—Dynamic programming (DP), global optimality, orthoimage, seamline detection, Voronoi diagram.**

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I. Introduction

SINGLE-SCENE orthophotos offer only limited spatial coverage and thus cannot satisfy the requirements of large-scale, continuous surface observation. Consequently, it is necessary to mosaic multiple overlapping images to produce seamless, wide-area remote sensing products, with seamline detection being the critical step that governs overall mosaic quality.

To balance computational efficiency and seamline quality, auxiliary priors are often introduced to constrain the search space. Existing prior-driven approaches can be broadly grouped into three categories: 1) *Elevation-Aware Methods:* these exploit digital surface models (DSMs) or disparity maps to steer seamlines away from abrupt elevation discontinuities. For instance, Zhang et al. [1] incorporated building heights into the cost function, while Pang [2] and Yuan et al. [3] generated dense disparity maps via SGM/SGBM to exclude tall obstacles, and Chen et al. [4] leverage DSM guidance to avoid above-ground structures; 2) *Vector-Map-Guided Methods:* these shrink the search domain using existing cartographic data. Wang et al. [5] derived medial axes between buildings to prevent seamlines from cutting through structures, whereas Wan et al. [6] constructed a weighted road-graph and solve a shortest-path problem so that seamlines preferentially follow the road network; and 3) *Semantics-Driven Methods:* these approaches leverage semantic information—such as object classes or probability maps—to guide seamline placement and avoid visually salient features. Yuan et al. [7] employed D-LinkNet to generate a road-probability map, encouraging seamlines to follow spectrally and texturally homogeneous road surfaces, and Li et al. [8] performed foreground segmentation in the overlap region to ensure seamlines bypass salient objects. Other priors include image classification results and overlap-area change detection [9], [10], [11], [12], [13]. However, DSMs, disparity maps, vector layers, and deep-network predictions are often unavailable or inconsistent across large, heterogeneous scenes. Prior-free, pixel-level approaches therefore remain advantageous: relying solely on the spectral, textural, and geometric characteristics of the imagery, they exhibit strong cross-regional robustness.

Current prior-free, pixel-level seamline-extraction algorithms can be broadly grouped into four categories:

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1) *Graph-Search Shortest-Path Methods:* Dijkstra [14], [15] and A* [16], [17] reformulate seamline detection as a shortest-path problem on a cost graph, yielding globally optimal solutions. However, their complexity rarely falls below $O(\mathrm{NM}\log(\mathrm{NM}))$, making them computationally prohibitive for tens-of-Mpixel imagery [18]; 2) *Global Optimization Approaches:* graph-cut frameworks [19], [20], [21] cast seamline selection as an energy-minimization problem, achieving higher seam quality than local search. Nevertheless, the computational complexity is $O(\mathrm{NM}(NM)^{1/2})$; even with tenfold image down-sampling [22], efficient processing of large scenes remains challenging; 3) *Iterative Optimization Methods:* ant-colony optimization (ACO) [23] and active contours (Snake models) [24] refine seamlines through iterative convergence, but incur high computational cost, are sensitive to parameter settings, and tend to become trapped in local minima, performing well only in limited scenarios; and 4) *Dynamic Programming (DP):* with linear complexity $O(\mathrm{NM})$, DP rapidly finds the minimum-cost path and significantly improves mosaicking throughput over large areas [25], [26]. However, its fixed subproblem decomposition and single propagation direction limit adaptability in complex terrain and high-frequency textures. To address these issues, this study extends DP by introducing multidirectional traversal and a "shrink-then-refine" strategy: the search space is first reduced by 90%, and the seamline is then refined within a narrow buffer using direction-flexible Dijkstra, achieving an optimal trade-off between efficiency and accuracy. Existing seamline-quality assessments rely mainly on subjective measures (e.g., counting the number of intersected buildings) or sparse metrics, such as RMS error [27], SSIM [28], and control-point registration error [17], which are insufficient to capture subpixel misalignment. To overcome these limitations, we propose a fully automated, pixel-level quality metric based on SIFT descriptors: within each seamline keypoint neighborhood, a 128-D gradient-orientation histogram is constructed, and after L_2 normalization and threshold truncation, local geometric continuity is sensitively quantified via Euclidean distance [29], enabling objective and repeatable evaluation.

1) We introduce a DP algorithm with enhanced search directions that steers the coarse seam-line away from texture-dense regions, and then apply pixel-level refinement within a narrow buffer zone—thereby achieving an optimal trade-off between processing speed and detection accuracy.
2) We develop a per-pixel SIFT-descriptor Euclidean-distance metric along the detected seam-line to quantify local geometric misalignment, effectively eliminating the subjectivity of traditional visual-inspection approaches.

The remainder of this article is organized as follows. Section II reviews related work. Section III presents the proposed methodology in detail. Section IV describes the experimental setup, comparative evaluations, and result analysis. Finally, Section V concludes this article.

II. Related Works

Seamline detection approaches in remote sensing image mosaicking can be broadly classified into two categories. Auxiliary-data-driven methods leverage external geospatial information—such as DSMs, road and building vector data, and LiDAR point clouds—to inform seamline placement, thereby avoiding elevated structures (e.g., buildings) and eliminating geometric misalignment of the same feature across image boundaries. Image-based methods, by contrast, rely exclusively on intrinsic image cues—utilizing color distributions or extracting object-level features—to steer seamlines clear of regions with pronounced grayscale disparities, thus minimizing visual discontinuities along the seams.

1) Auxiliary-data-driven seamline methods primarily rely on external geospatial information—such as DSMs, road and building vector data, or LiDAR point clouds—to guide seamline placement. Ma and Sun [30] applied the A* algorithm to point-cloud data to avoid obstacle regions, but nonground returns in mountainous scenes lead to instability. Wang et al. [5] used geometric angle bisectors between adjacent buildings to route seamlines around building perimeters. Wan et al. [6] constructed a cost map from road vectors and employed the Floyd–Warshall algorithm [31] to compute the shortest paths, thereby constraining seamlines to road corridors. Chen et al. [32] leveraged DSMs to steer seamlines toward low-elevation areas and proposed an orthophoto elevation synchronization model (OESM), using morphological processing of OESM data to generate a seamline network. Zheng et al. [16], [33], [34] built an initial Voronoi network from DSM-derived edge maps [15], [35] and relocated vertices to avoid prominent features, achieving seamless mosaicking. Zhang et al. [1] incorporated approximate object heights estimated from DSMs as weighted terms into traditional gradient-and-grayscale cost functions and solved for the optimal seamline via graph cuts. Varshosaz et al. [36] first generated a DSM and overlaid a low-resolution grid, selecting the optimal image for each grid cell to construct a seamline network, thereby reducing artifacts and discontinuities in overlapping regions. Yoon and Kim [37] proposed an autonomous aerial vehicle (AAV) image stitching technique based on a triangulated irregular network (TIN) constructed from sparse ground points; closely related TIN strategies further indicate that triangulations built from tiepoints can suppress relief-induced mismatches while maintaining minute-level runtimes [38], [39], [40]. Overall, auxiliary-data-driven methods tend to perform better when high-quality external data are available over small areas; however, their dependence on data coverage and precision limits applicability to large-scale mosaicking tasks.

2) Image-based seamline methods leverage salient object-level features or pixel-level cost information and can be divided into two categories. The first comprises object-level feature extraction approaches: Pan et al. [12] performed object-level change detection by measuring the proportion of changed pixels within segmented regions to prevent seams from crossing significantly changed areas; Yuan et al. [41] fused dense optical flow with disparity-map optimization to automatically enhance building outlines and roof structures for DOM

generation; Pang et al. [2] and Yuan et al. [3] extracted disparity via SGM and SGBM, respectively, to reduce search nodes; Wang et al. [9] located seams using watershed segmentation and regional similarity; Yuan et al. [7] produced road-probability maps with a D-LinkNet model to guide seams along streets and avoid major buildings; Zhang et al. [42] coupled SIFT with DP under an HVS-guided strategy to make seams perceptually unobtrusive; Liu et al. [43] proposed a segmentation-driven SOSG pipeline for SAR mosaicking that builds a Dijkstra cost from homogeneous regions and intensity differences; and Wang et al. [44] adopted superpixel-based multilabel optimization to determine superpixel-level optimal seamlines across multi-image overlaps. The second category comprises raster-level cost-minimization methods, which construct pixel-wise cost surfaces from color, gradient, or texture cues and then search for minimum-cost paths: Kerschner [24] proposed the Dual Snake algorithm, which evolves contours from both ends on a color–texture cost map but is prone to local minima; Zhang et al. [45] and Wang et al. [23] augmented ACO with gray/gradient constraints and Max–Min rules to improve global convergence; Davis et al. [18] applied Dijkstra's algorithm to mitigate stitching errors from lens distortion; Chon et al. [46] used normalized cross correlation as the cost metric; Lin et al. [47] iteratively shrinked the feasible search region before using Dijkstra to minimize the maximum cost; Li et al. [21], [28], [48], [49], [50], [51] widely employed graph-cut energy minimization, whose exponential computational cost is alleviated by Chai et al. [22] via progressive downsampling and buffered refinement; a two-stage seamline search combined with homography-based detail-aware matching improves robustness to local deformations [52]; Pan et al. [11] integrated multiregion segmentation with graph cuts for AAV mosaicking; Peng et al. [53] introduced feature matching and elastic deformation prior to graph cuts for high-texture AAV hyperspectral stitching; Li's APDP and Shen's multidirectional SMP-DP offer efficient DP searches but are limited by direction count [25], [26]; Pan et al. [54] adopted the fast marching method to reduce ghosting, misalignment, and color artifacts; quaternion-domain perceptual modeling jointly optimizes color-aware seamlines and performs iterative local alignment [55]; Gao et al. [56] coupled BRISK+GMS registration with optimal-seam fusion for real-time scenarios; and Li et al. [57] used weighted fast sweeping for globally optimal seam detection with seam-constrained optical-flow alignment to suppress residual parallax. Complementary to seamline detection, coarse-to-fine trust-region image warping can be invoked along detected seamlines—and extended to multi-image joint optimization—to correct geometric misalignments that seam placement alone cannot resolve [58].

The integration of path-search algorithms with image feature fusion has become the predominant approach for automatic seamline extraction. However, when applied to large-scale, multiterrain, and cross-scene remote sensing imagery, it still exhibits significant limitations. First, acquiring exogenous data—such as high-resolution DSMs or building and road vector layers—is costly and often lacks sufficient spatial coverage, impeding large-scale adoption. Second, deep semantic or disparity models frequently suffer from feature mismatches and poor cross-domain generalization when transferred among complex land-cover types such as urban, mountainous, and agricultural areas. Finally, grid-based path searches (e.g., DP, ACO, graph cuts) on imagery containing hundreds of millions of pixels incur nonlinearly increasing computational and storage demands, making it difficult to achieve both efficiency and global optimality. Therefore, devising a unified framework that operates at the pixel level without relying on external data—while still ensuring high efficiency and precision—remains a critical challenge for improving the stability, scalability, and robustness of seamline detection in practical production environments.

III. Methods

Seamline detection in remote sensing image mosaicking must satisfy both geometric and radiometric requirements. Geometrically, the seamline should avoid crossing prominent object boundaries, such as buildings or roads. Radiometrically, it should minimize color or brightness differences across the seam. This task can be abstracted as a cost-optimal path search within the overlapping region of adjacent images, where the cost function must jointly penalize geometric misalignments and radiometric inconsistencies. To achieve this, we propose a three-stage processing framework, as illustrated in Fig. 1. First, the input Level-1 satellite imagery undergoes orthorectification [59], [60], [61] and color correction to produce geometrically and radiometrically consistent Level-3 products [62], [63], [64], [65], [66], [67], [68] serving as the foundation for subsequent high-precision processing. Next, based on the overlap between image pairs, an initial seamline network is automatically constructed to provide a topological basis for global mosaicking and constrain the subsequent path search. Then, a two-step seamline detection strategy is employed to extract high-quality seamlines: in the first step, a five-directional DP algorithm, guided by a cost function incorporating entropy and gradient information, rapidly identifies a coarse seamline that avoids major features. In the second step, a narrow buffer zone is constructed along the coarse seamline, within which an enhanced Dijkstra algorithm performs pixel-level refinement, enabling fast and accurate detouring around complex objects. Finally, the images are cropped and mosaicked according to the detected seamlines, resulting in a geometrically seamless and radiometrically smooth wide-area orthophoto mosaic.

A. Initial Seam-Line Network Generation for Batch Seam-Line Detection

Existing pairwise seam-line methods compute boundaries only between two adjacent orthophotos and then assemble mosaics sequentially in input order, resulting in repeated disk I/O and suboptimal efficiency. To overcome this limitation, it is preferable to initialize a global seam-line network across all input images in a single operation. Pan et al. [15], [35] introduced the AVDO (Overlapping-Region Voronoi Diagram) method, which constrains Voronoi edges to each pairwise overlap and places vertices within the common overlap of the entire image set. Similarly, Yuan et al. [69] detected boundary vectors of valid orthophoto extents, construct the

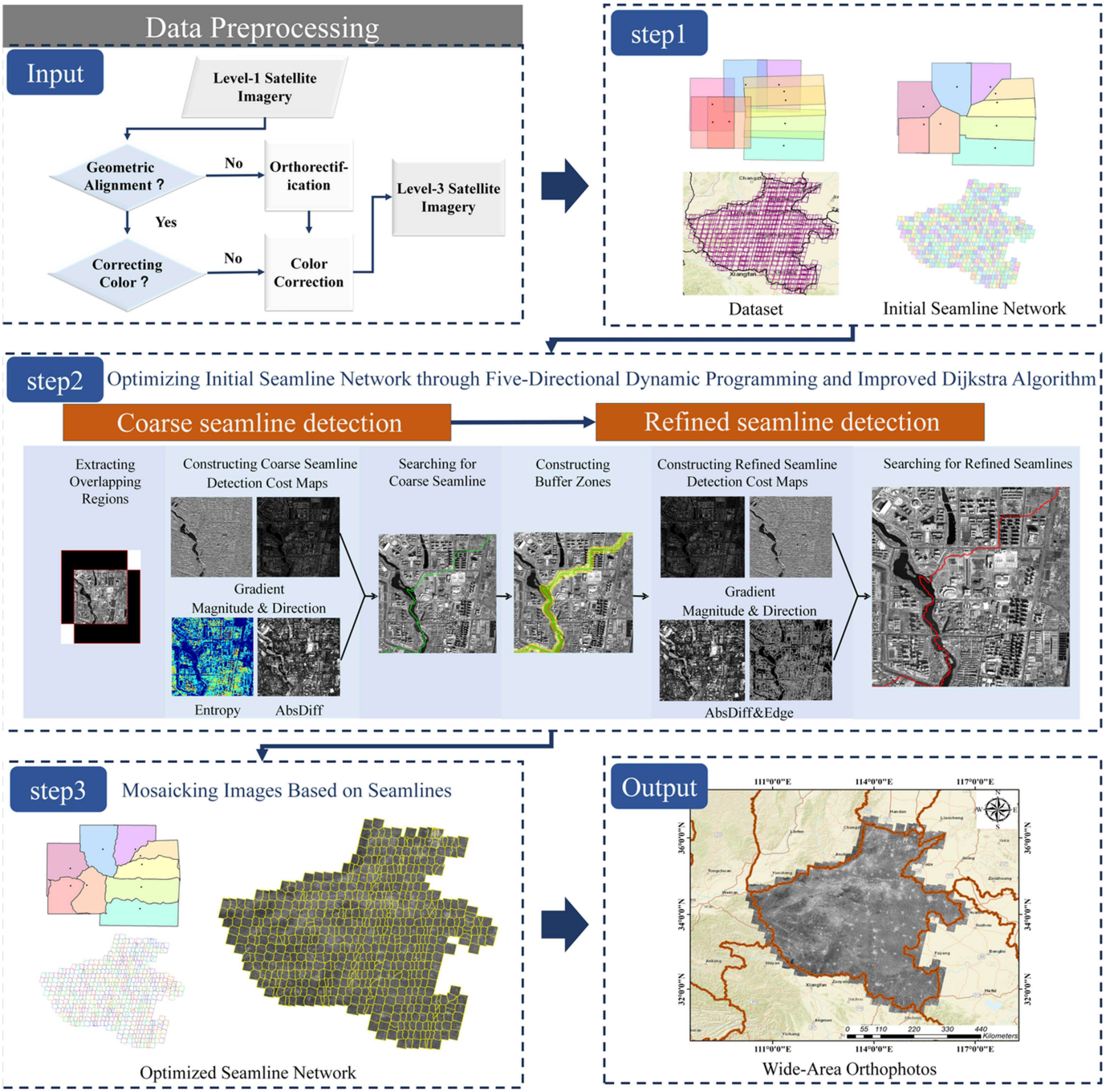

Fig. 1. Framework of the proposed method.

corresponding Voronoi diagram, and extract its medial axes as initial seam-lines. Thereafter, a pixel-level detection algorithm is applied to refine the entire network, yielding a globally optimized seam-line configuration. Crucially, this approach requires each image to be read and written only once, markedly enhancing I/O efficiency.

The proposed seam-line detection framework necessitates an initial network that conforms to 2-D DP constraints. Accordingly, we generate this network by iteratively clipping a global Voronoi diagram with each orthophoto's valid range polygon (VRP) [70]. First, the VRP of each image is extracted, and its exterior boundary is defined as the clipping domain. Using the geometric centroids of all orthophotos as Voronoi sites, a global Voronoi tessellation is computed; the intersection of each cell with its corresponding VRP yields that image's effective mosaic polygon (EMP). Any residual polygon lying inside an image's VRP but outside its EMP is then transferred to the EMP of the adjacent image. This clipping-and-merging process repeats across all images until every EMP is entirely contained within its own VRP. The result is a repartitioned Voronoi diagram whose edges—each representing a seam-line segment—either coincide with image boundaries or lie within overlap regions. Each segment is subsequently used as the principal diagonal of a rectangular extraction window,

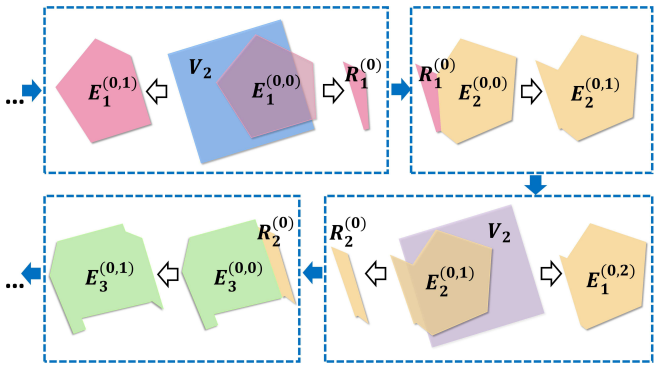

Fig. 2. Illustration of the iterative clipping process between VRPs and EMPs.

allowing small image tiles to be cropped and thereby reducing the scale of the cost map for the downstream DP search. The detailed process is as follows [see Fig. 2].

First, for each orthoimage i, the valid coverage polygon VRP_i is extracted by excluding regions with NoData values. This polygon precisely delineates the valid pixel area of the orthoimage. Then, the union of all VRP_i polygons

$$\Omega = \bigcup_{i=1}^{n} \mathrm{VRP}_i \tag{1}$$

is computed as the global clipping region Ω, ensuring that the subsequently generated Voronoi cells do not extend beyond the valid coverage area of any image.

The geometric centroid C_i of each orthoimage is selected as the seed point for generating the Voronoi cells. A standard 2-D Voronoi diagram is then constructed within the global clipping region Ω, and the Voronoi cell corresponding to the ith image is denoted as V_i. This cell is further clipped by the union region to obtain

$$\mathrm{EMP}_i^{(0,0)} = V_i \cap \Omega. \tag{2}$$

In this way, each image obtains an EMP.

Define the Voronoi adjacency set $N(i)$ and represent it as an ordered sequence $\{j_1, \ldots, j_{|N(i)|}\}$

$$N(i) = \left\{ j \neq i \mid V_i \cap V_j \neq \emptyset \right\}, \quad N(i) = \left\{ j_1, \ldots, j_{|N(i)|} \right\}. \tag{3}$$

Next, before the start of the tth iteration, the current $\mathrm{EMP}_i^{(t,0)}$ of image i is computed, and its difference with its own VRP is taken as

$$\mathcal{R}_i^{(t)} = \mathrm{EMP}_i^{(t,0)} \setminus \mathrm{VRP}_i. \tag{4}$$

Let $\mathcal{R}_i^{(t)} = \{R_{i,1}^{(t)}, R_{i,2}^{(t)}, \ldots, R_{i,m_i}^{(t)}\}, R_{i,p}^{(t)} \subset \mathbb{R}^2$. denote the residual set, which contains all polygons located within the Voronoi cell but falling outside the valid range VRP. Let the sequence of Voronoi cells adjacent to seed point i be $\{j_1, \ldots, j_{|N(i)|}\}$. Following this order, for each substep $k = 1, \ldots, |N(i)|$, all residual polygons of image j_k are sequentially merged into its own EMP

$$\mathrm{EMP}_i^{(t,k)} = \mathrm{EMP}_i^{(t,k-1)} \cup \bigcup_{p=1}^{|\mathcal{R}_{j_k}^{(t)}|} R_{j_k,p}^{(t)}, k = 1, 2, \ldots, |N(i)|. \tag{5}$$

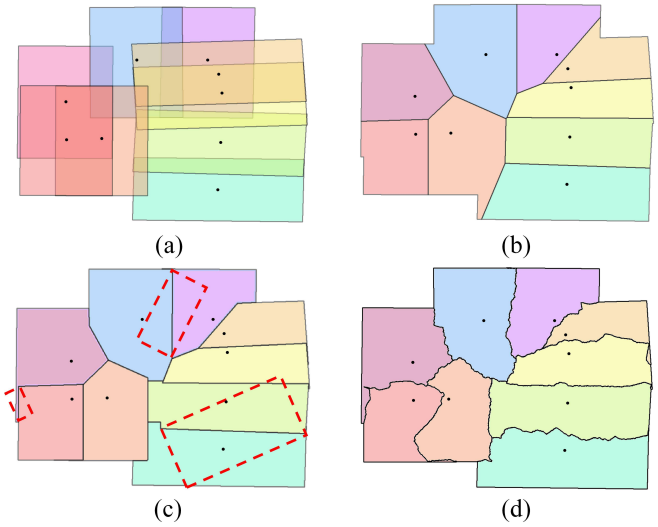

Fig. 3. Construction and optimization of the initial seamline. (a) VRP of the image. (b) Voronoi diagram of the image. (c) Initial seamline network with local image selection. (d) Optimized seamline network.

After completing all $|N(i)|$ merge operations, the initial EMP for the next iteration is obtained as

$$\mathrm{EMP}_i^{(t+1,0)} = \mathrm{EMP}_i^{(t,|N(i)|)}. \tag{6}$$

This process is repeated until, and only until, in a certain iteration, no new residual polygons are generated for any image, that is

$$\forall i: \ \mathcal{R}_i^{(t)} = \emptyset \Leftrightarrow \mathrm{EMP}_i^{(t,0)} \subseteq \mathrm{VRP}_i. \tag{7}$$

At this point, the iteration naturally terminates. The global Voronoi diagram is continuously repartitioned through repeated clipping and assignment, ultimately producing an EMP for each image. After the iteration completes, the boundaries of these EMPs serve as the initial seamlines. These boundaries preserve the geometrically balanced partitioning property of the Voronoi diagram while ensuring that each seamline segment either lies on the image boundary or within the overlapping area between adjacent images [see Fig. 3(b)], thus providing a solid foundation for subsequent seamline refinement.

After obtaining the initial seamline network, each initial line segment is first used to define a circle on the plane, with its two endpoints serving as the diameter endpoints. According to the inscribed angle theorem, any point on the circle that connects to the diameter endpoints forms a right triangle. Leveraging this geometric property, and using the straight line containing the diameter as a reference, two perpendicular directions from the endpoints to the circle's arc are used to determine the minimal bounding rectangle that contains the segment [see Fig. 3(c)]. Next, a local subimage is extracted from the original orthoimages by cropping the pixel region corresponding to this bounding rectangle. On this local image, the previously described seamline search algorithm (the two-stage method combining DP and Dijkstra) is applied to recompute the locally optimal path. For regions outside the VRP, a large cost is assigned to treat them as impassable areas. Once a refined

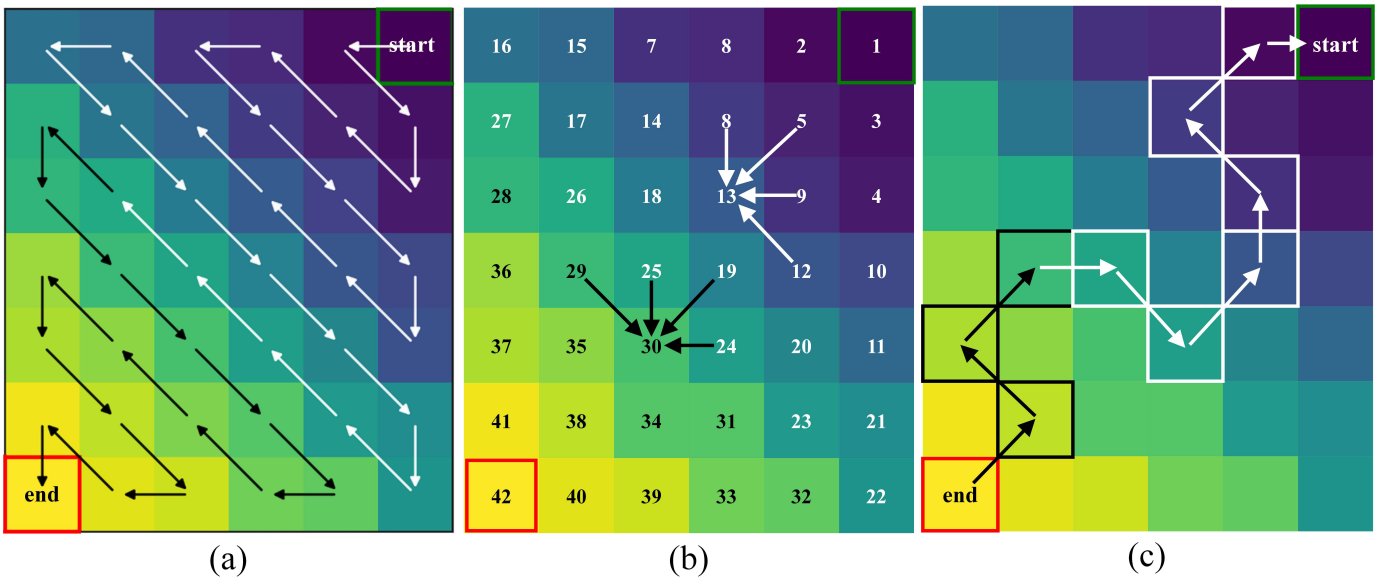

Fig. 4. Illustration of the improved directional DP traversal and backtracking process. (a) Traversal order of the DP algorithm. (b) Cost accumulation directions in the DP algorithm. (c) DP backtracking process.

path is successfully computed, it replaces the original initial segment. This process is repeated for every segment in the initial seamline network until all segments have been replaced by locally optimized paths, thereby completing the refinement of the initial seamline network [see Fig. 3(d)].

B. Dynamic Programming-Based Coarse Seamline Detection

To find a "minimum-cost path" between two images A and B, it is usually necessary to construct a grid (or a set of nodes), where each grid node contains information from the corresponding position on the two images; define a path from the starting point to the ending point, and compute its "overall cost" on the images; then apply the DP method to search for the optimal path based on a certain cost accumulation strategy. The DP algorithm can find a globally optimal path with the minimum cost. It can be described as follows: based on image information, an n $\times$ m 2-D matrix is constructed. The cost of the initial grid cell is calculated, and according to a predefined traversal order, the cost of each grid cell is computed. Once the cost of each cell is determined, the solution of each subproblem is also determined. Then, starting from the endpoint, a backtracking process is performed along the predefined directions. Each step of the backtracking corresponds to the optimal solution of a subproblem. When the start point is reached, the minimum-cost path is obtained. This article formalizes the seamline search problem as a 2-D DP process based on a regular grid. The process accumulates local costs, performs step-by-step reasoning in an acyclic topological structure, and ultimately derives a path with the minimum total cost. The specific method includes three core components: traversal strategy design, cost function construction, and state transition equation.

1) Traversal Strategy Design: To accommodate the diversity and directional variation of object edge information in complex scenes, this article introduces a directional control strategy into the DP path search process, aiming to enhance the flexibility and robustness of path propagation in irregular structures. Traditional DP algorithms often adopt fixed adjacent directions, which can lead to rigid backtracking paths and even result in information overlap or path loops. To address this, a loop-free five-directional mechanism based on anti-diagonal encoding is designed in this study. Specifically, each pixel in the image is assigned an anti-diagonal index based on the sum of its coordinates $s = i + j$, and two directional propagation sets are defined according to the parity of s. Here, i denotes the row index from top to bottom, i.e., the vertical index, and j denotes the column index from left to right, i.e., the horizontal index [see Fig. 4].

When s is even (i.e., on an even-numbered anti-diagonal), the following direction set is used:

$$\mathcal{P}_{\text{even}} = \{(-1, 0), (-1, +1), (0, +1), (+1, +1)\} \tag{8}$$

which allows transitions from the top, top-right, right, and bottom-right directions.

When s is odd (i.e., on an odd-numbered anti-diagonal), the following direction set is used:

$$\mathcal{P}_{\text{odd}} = \{(-1, -1), (-1, 0), (0, +1), (+1, -1)\} \tag{9}$$

which allows transitions from the top-left, top, right, and bottom-left directions.

The alternating use of these two direction sets ensures that the state of each pixel depends only on the current and preceding anti-diagonals, thereby forming a strictly acyclic topological structure (DAG) that guarantees both the uniqueness of the path traversal order and the unidirectionality of information propagation. This structure not only enhances the directional adaptability of the path but also effectively prevents local backtracking and conflicting updates, making it particularly suitable for single-source shortest path search tasks from the image's starting point to the endpoint.

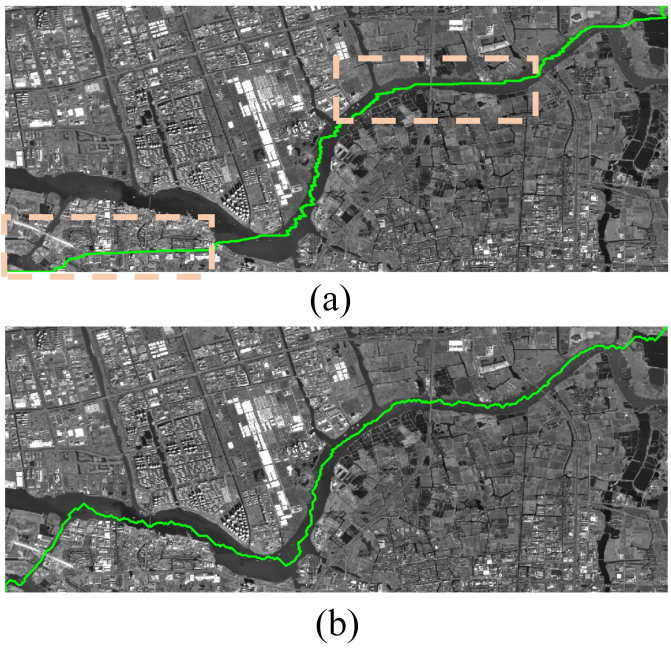

Fig. 5. (a) Seamline detected by the SMP-DP method [26]. (b) Seamline detected by the proposed method.

Fig. 5 compares the seam-line extraction results of the SMP-DP method and the proposed approach on the same pair of remote-sensing images. Fig. 5(a) shows the output of SMP-DP [26], while Fig. 5(b) depicts the seam line obtained by our method. It can be seen that, in regions containing river bends, SMP-DP—restricted to its single, predefined search direction—cannot search downward and thus fails to traverse these complex features, causing its seam line to deviate from the true optimal path. In contrast, the path produced by our method conforms more closely to the underlying feature boundaries: it demonstrates greater local responsiveness and directional flexibility, particularly around river meanders and in densely parceled areas.

The proposed multidirectional DP algorithm evaluates candidate predecessors along four principal propagation vectors—primary diagonal, secondary diagonal, horizontal, and vertical—at each pixel node, and selects the globally minimal, cycle-free transition to construct the optimal subpath. This directional flexibility enables the seam line to smoothly circumvent complex features—such as river bends and road corners—by dynamically switching propagation direction at points of curvature discontinuity. In contrast, SMP-DP confines its search to a small, fixed set of directions: although it guarantees optimality within those directions, it cannot adapt to arbitrary local curves, resulting in rigid, piecewise linear deviations that fail to conform to true feature geometry at bends and corners. By leveraging a richer set of propagation directions, our method achieves markedly improved local curve fitting and alignment.

2) Cost Function: The first stage of the proposed method aims to generate an initial cost map that accurately reflects structural differences between images by combining image downsampling and a structure-sensitive gradient-based cost function. Considering the complexity and high computational cost of full-resolution images, we first downsample the input overlapping grayscale images $I1$ and $I2$, and construct a pixel-wise gradient-structure-based cost map $C(i, j)$. This cost integrates four components: grayscale difference, gradient magnitude similarity, gradient direction difference, and local entropy.

For the images $I1$ and $I2$, at any grid point (i, j), we define its cost $C(i, j)$ as the weighted sum of the following four terms:

$$C(i,j) = \lambda_d C_d(i,j) + \lambda_m C_m(i,j) + \lambda_g C_g(i,j) + \lambda_h C_h(i,j) \tag{10}$$

where $C_d(i, j)$ represents the grayscale difference between images, measuring the local intensity deviation between corresponding pixels. It is used to capture local radiometric inconsistencies between the two images. In practice, grayscale deviations may arise due to varying imaging conditions, such as sensor sensitivity, exposure time, atmospheric conditions, or surface reflectance. If such differences are not properly addressed, the seamline may be biased toward regions with abrupt intensity changes. By directly incorporating brightness difference as a cost component, pixels with large interimage discrepancies are penalized, encouraging the final path to avoid high grayscale difference regions and preferentially follow areas with similar brightness

$$C_d(i,j) = \frac{\left| I_1(i,j) - I_2(i,j) \right|}{I_{\max} - I_{\min}}. \tag{11}$$

Here, $I_1(i, j)$ and $I_2(i, j)$ denote the grayscale values of the two overlapping images at position (i, j), and $I_{\max}, I_{\min}$ represent the theoretical maximum and minimum intensity values of the images, ensuring $C_d(i, j) \in [0, 1]$.

$Cm(i, j)$ represents the gradient magnitude similarity term. In remote sensing image mosaicking, an ideal seamline should avoid crossing the edges of real-world objects—such as roads, rivers, and building contours—as cutting through these structured features often leads to noticeable geometric discontinuities and visual artifacts along the seam. Gradient magnitude is a classical measure of the intensity of brightness changes in an image: it tends to peak at object boundaries and remains low in large homogeneous regions, such as farmland, grassland, bare soil, or water surfaces. When both images exhibit strong gradients at the same location, it typically indicates the presence of a real object boundary (e.g., a road or riverbank). In such cases, the use of a minimum operator results in a high value for $Cm(i, j)$, increasing the cost at that location and guiding the algorithm to bypass it. The gradient-based cost term is defined as

$$C_m(i,j) = \frac{\min\left(\|\nabla I_1(i,j)\|, \|\nabla I_2(i,j)\|\right)}{\nabla I_{\max}}. \tag{12}$$

Here, $\|\nabla I_k(i, j)\|$ denotes the gradient magnitude of image k at pixel (i, j), and $\nabla I_{\max}$ is the theoretical maximum gradient magnitude determined by the chosen gradient operator and intensity range.

$C_g(i, j)$ denotes the gradient direction difference term, which measures the consistency of gradient orientation between the two images. The purpose of introducing this term is to

enhance the algorithm's ability to recognize direction-sensitive structures, such as slanted edges or texture boundaries. This term is computed by summing the differences in first-order gradients along the horizontal and vertical directions, which characterizes whether the texture orientation at a given pixel is aligned between the two images. A large difference in edge directions across the two sides indicates a high likelihood of visual discontinuity or angular mismatch if the seamline passes through that location. Conversely, if the gradient directions are nearly aligned, it suggests a coherent extension of the same object across both images. When the path encounters a region with a significant directional mismatch, it is penalized, thereby guiding the seamline toward object boundaries with consistent orientation and structural continuity. The directional consistency cost is defined as

$$C_g(i,j) = \frac{\left|\theta_1(i,j) - \theta_2(i,j)\right|}{\pi} \tag{13}$$

where $\theta_1(i,j) = \arctan 2(G_{y1}(i,j), G_{x1}(i,j))$ and $\theta_2(i,j) = \arctan 2(G_{y2}(i,j), G_{x2}(i,j))$ are the gradient orientations at pixel (i,j) in the first and second image, respectively. This normalization makes $C_g(i,j) = 0$ when the orientations are identical and $C_g(i,j) = 1$ when they differ by 180°.

$C_h(i,j)$ represents the texture complexity penalty term based on local image entropy, which is used to discourage the path from passing through texture-rich regions. To construct the local entropy cost map, a sliding window is applied centered on each pixel, and the entropy within the window is computed as a metric for local texture complexity. The computation of local entropy can be parallelized using OpenMP for improved efficiency.

Entropy is calculated based on the grayscale histogram of the window, essentially measuring the uncertainty and complexity of the gray-level distribution in the region. For each image patch of size $w \times h$ (with $w = h = 10$), the grayscale values are uniformly quantized into L levels (set to $L = 32$ in this article). The number of pixels in each gray level forms the local grayscale histogram. Let n_k denote the number of pixels at gray level k, and the total number of pixels be $N = w \cdot h$. Then, the frequency of gray level k is given by

$$p_k = \frac{n_k}{\mathrm{wh}}. \tag{14}$$

The entropy value H of the local patch is calculated as follows:

$$H = -\sum_{k=1}^{L} p_k \log_2 p_k. \tag{15}$$

To enhance the discriminative power of entropy values within the cost map, a nonlinear enhancement is applied to the entropy values using a cubic function, which increases the penalty strength in high-entropy regions. Finally, the cost value assigned to the corresponding region of the image patch is defined as

$$C_h(i,j) = \begin{cases} 0, & H \le \tau \\ H^3, & H > \tau. \end{cases} \tag{16}$$

The threshold τ is set to 2.8 to distinguish between low-texture regions and structurally complex areas. The computed

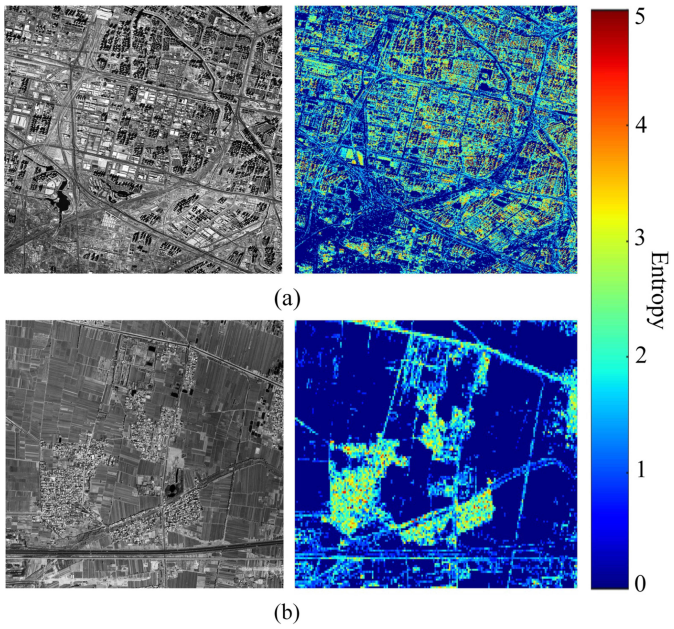

Fig. 6. Local entropy computation results. (a) Densely built urban area. (b) Sparsely built village area.

$C_h(i,j)$ is assigned to all pixel positions within the corresponding image patch, thereby forming a local entropy cost map that is consistent in size with the original image. This entropy map will be used in the subsequent path search or cost fusion process as an explicit structural complexity penalty term, and will be superimposed onto the base cost map [see Fig. 6].

To ensure that the cost map simultaneously suppresses radiometric discrepancies and avoids structurally complex regions, all four components are normalized to $[0,1]$ (for the entropy term, since $H \in [0, \log_2 L]$, we apply cubic enhancement and then normalize by $(\log_2 L)^3$), and we adopt the weights $\lambda_d = 2$, $\lambda_m = \lambda_g = 1$, and $\lambda_h = 5$. The rationale is as follows. First, assigning the largest weight to the entropy term ($\lambda_h = 5$) is motivated by perceptual considerations and robustness: high-entropy areas (e.g., dense urban textures, forests, and ridge patterns) are most prone to visible artifacts along the seam, and their uncertainty persists even after downsampling. A strong penalty (cubic enhancement plus a large weight) markedly reduces the probability that the seam traverses such regions, while the threshold τ zeroes out large, low-texture areas and prevents raising the background cost. Second, $\lambda_d = 2$ compensates for the prevalence of radiometric differences and the typically lower variance of C_d in real data: although $C_d \in [0,1]$ often concentrates in the lowe–middle range, illumination/sensor inconsistencies can directly induce visible seams. A moderately larger weight for C_d prioritizes avoidance of strong brightness discontinuities without unduly sacrificing geometric continuity. Third, setting $\lambda_m = \lambda_g = 1$ reflects equal emphasis on edge strength and edge-direction consistency: relying solely on magnitude may treat homologous boundaries as obstacles, whereas relying solely on orientation can fail in weak-texture regions. Equal weighting balances "is there a true boundary"' (magnitude) and "are structures aligned across images"' (orientation), reducing bias. Overall, this

configuration is aligned with the empirical distributions of the terms, the noise characteristics after downsampling, and human visual sensitivity to texture and edges: the strong entropy penalty preserves perceptual quality, the boosted radiometric term suppresses brightness seams, and the equally weighted gradient terms maintain geometric coherence. In diverse terrains (urban/mountain/agricultural) and multisource datasets, this setting has shown stable performance; in extreme cases—e.g., severe illumination mismatch or exceptionally flat texture—it can be gently adapted following the same principles (e.g., increasing λ_d or slightly decreasing λ_h).

3) State Transition Equation: Let $D(i, j)$ denote the minimum cumulative cost from the predefined starting point [e.g., the top-right corner $(0, W-1)$] to the current point (i, j). Its state transition recurrence relation is defined as follows:

$$D(i,j) = \begin{cases} C(i,j), & \text{if } (i,j) = (0,\ W-1) \\ \min\limits_{(\Delta i,\Delta j)\in\mathcal{P}_{s \bmod 2}} \left[D(i+\Delta i,\ j \right. \\ \qquad \left. +\Delta j) + C(i,j) \right], & \text{otherwise} \end{cases} \tag{17}$$

where $C(i, j)$ denotes the cost at pixel (i, j). $s = i + j$ represents the anti-diagonal index on which the current pixel lies, $\mathcal{P}_{s,\,\mathrm{mod}\,2}$ denotes the set of predecessor directions used on that anti-diagonal, and $(i+\Delta i, j+\Delta j)$ represents a valid predecessor node. All invalid indices (e.g., those outside the image boundary) must be skipped during implementation. This state transition mechanism ensures that each pixel's optimal cumulative cost depends only on its reachable predecessors and is updated by minimizing the local cost function. Combined with the Zi–Zag anti-diagonal scanning strategy, the method efficiently extracts the globally optimal path with time complexity $O(\mathrm{HW})$.

Once the cumulative cost matrix $D(i, j)$ has been completely filled, a coarse seamline can be traced by backtracking from the terminal node (typically chosen as the lower-left corner $(H-1, 0)$).

Let $\mathbf{P} = \{(i_k,\ j_k) \mid k = 0,\ \ldots,\ K\}$ denote the sequence of nodes obtained during backtracking. For the current node (ik, jk), define its anti-diagonal index as $sk = ik + jk$. If $(ik, jk) = (0, W-1)$, backtracking terminates; otherwise, from the direction set $\mathcal{P}s_k, \mathrm{mod}, 2$ select the predecessor that yields the minimum cumulative cost

$$\left(i_{k+1}, j_{k+1}\right) = \min_{(\Delta i,\Delta j)\in\mathcal{P}_{s_k\ \mathrm{mod}\ 2}} D(i_k+\Delta i,\ j_k+\Delta j). \tag{18}$$

By iterating in this manner, a sequence of backtracking path nodes is formed. The process is repeated until the starting point is reached, after which the sequence $\mathbf{P}$ is reversed and output as the coarse seamline $\mathbf{S}_{\mathrm{coarse}}$.

C. Dijkstra Algorithm Within the Buffer Zone for Pixel-Level Seamline Refinement

While the coarse seamline **S**coarse ensures global optimality, a finer seamline can be achieved by leveraging the directional flexibility of the Dijkstra algorithm, which supports eight-directional search in a 2-D matrix. To this end, we propose a pixel-level Dijkstra-based refinement strategy constrained by a buffer zone, which utilizes spatial priors from the coarse path to significantly reduce the search space and perform high-resolution local cost optimization, resulting in the final fine seamline $\mathbf{S}_{\mathrm{fine}}$.

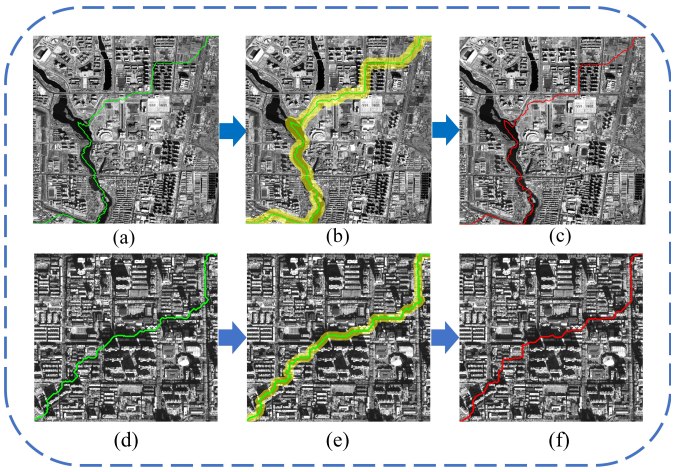

Fig. 7. Illustration of local buffer zone construction and refined search. (a) and (d) Coarse seamlines detected by the DP algorithm. (b) and (e) Local buffer zones. (c) and (f) Refined seamlines detected by the Dijkstra algorithm.

Specifically, a buffer region is constructed along the coarse seamline, with its width determined based on application requirements and image resolution—typically set to 25 pixels [see Fig. 7]. The node set within the buffer includes only those pixels located within a limited vicinity of the coarse seamline, effectively constraining the number of candidate nodes and improving computational efficiency.

Within this buffer zone, a Dijkstra-based fine search strategy is employed. The local cost for each node is defined based on factors such as grayscale difference, gradient magnitude, and edge information, to precisely capture local feature variation and structural detail. The nodes on the coarse seamline serve as the initialization points, from which the Dijkstra algorithm incrementally expands to cover all nodes within the buffer, dynamically updating the optimal path cost and extracting the minimum-cost fine seamline. Thanks to the spatial limitation imposed by the buffer, the Dijkstra algorithm can quickly cover all valid nodes without conducting an exhaustive pixel-wise search over the entire image. This dramatically reduces the number of expanded nodes and the overall computational burden. Through this two-stage coarse-to-fine framework, the proposed method achieves fast and accurate seamline refinement, delivering high-quality stitching results with strong robustness and efficiency in practical applications.

1) Optional Image Downsampling Within Overlapping Regions: Let the coarse path contain $K+1$ nodes $\mathbf{P} = \{(x_k, y_k); \mid; k = 0, \ldots, K\}$. In the original pixel coordinate system, a buffer mask $M \in 0, 1^{H\times W}$ is created by centering a radius-r band around $\mathbf{P}$

$$M(i,j) = \begin{cases} 1, & \min_k \sqrt{(i-y_k)^2 + (j-x_k)^2} \le r \\ 0, & \text{otherwise.} \end{cases} \tag{19}$$

Here, $M(i, j) = 1$ indicates that pixel (i, j) lies within the buffer zone, while $M(i, j) = 0$ means it is excluded from

further consideration. The buffer zone retains only pixels within narrow bands on either side of the coarse seamline, significantly reducing the total pixel count N compared to the original image (typically $N/\mathrm{HW} < 5\%$), thereby providing an exponential speed-up for the subsequent high-resolution search.

2) Cost Function: In the path refinement stage, to enhance the sensitivity of the seamline to true structural boundaries, a new pixel-level local cost function $C'(i, j)$ is defined within the buffer zone. It is formulated as a linear combination of three components: radiometric difference, gradient similarity, and a penalty term. Specifically

$$C'(i,j) = \lambda'_d C'_d(i,j) + \lambda'_m C'_m(i,j) + C'_p(i,j) \tag{20}$$

where $C'_d(i, j)$ is used to measure the grayscale difference between the two images at the same pixel location. It corresponds to the radiometric difference term in the coarse seamline stage and ensures that the path avoids regions with abrupt radiometric changes. $C'_m(i, j)$ represents the gradient magnitude similarity term, consistent with the edge-constraining strategy used in the coarse seamline stage. This term is designed to prevent the path from crossing real object boundaries, thereby guiding it through texture-homogeneous regions.

$C'_{\mathrm{p}}(i, j)$ is a penalty term designed to impose hard constraints on "over-threshold grayscale transitions" and "explicitly detected edges." Specifically, when the grayscale difference exceeds a predefined threshold τd or when an edge is detected by the Canny operator (i.e., $E(i, j) = 1$), a strong penalty is triggered. This term works in conjunction with the radiometric difference C'_{d} and the gradient magnitude similarity C'_{m}: the first two provide soft, continuous cost gradients to encourage avoidance of radiometric discontinuities and shared object boundaries, while C'_{p} enforces hard exclusion at locations with sharp grayscale transitions or explicit edges. This combination ensures that the seamline path avoids high-risk pixels as much as possible, thereby preserving overall smoothness while minimizing the likelihood of crossing object structures. This term is defined as follows:

$$C'_{\mathrm{p}}(i,j) = \beta \mathbf{1}\left(|I1(i,j) - I_2(i,j)| > \tau_d\right) + \gamma \mathbf{1}\left(E(i,j) = 1\right). \tag{21}$$

Here, τd is the grayscale discontinuity threshold, and $E(i, j)$ is obtained by applying the Canny operator to the absolute grayscale difference image. $\mathbf{1}$ denotes the indicator function. β and γ are large penalty coefficients, both set to 1000. λ'_d and λ'_m are the weights used in the refinement stage, where λ'_d is set to 2 and λ'_m is set to 1. In the formula, $\mathbf{1}(|I1(i, j) - I2(i, j)| > \tau d)$ means "this term equals 1 if the grayscale difference exceeds the threshold τd, otherwise it equals 0"; $\mathbf{1}(E(i, j) = 1)$ means "this term equals 1 if the pixel (i, j) is detected as an edge point in the edge map, otherwise it equals 0."

In the subgraph $G = (\mathcal{V}, \mathcal{E})$ defined by the buffer mask, each valid pixel (i, j) corresponds to a node, and the eight-neighborhood pixels form the undirected edge set $\mathcal{E}$. Let the cumulative cost map be denoted as $G(i, j)$; then the pixel-level shortest path satisfies the Bellman–Dijkstra recurrence

$$G(i,j) = \min_{(\Delta i, \Delta j) \in \mathcal{N}_8} \{G(i + \Delta i,\ j + \Delta j)\} + C'(i,j), \quad (i,j) \in \mathcal{V} \tag{22}$$

where the search directions are defined over an eight-connected neighborhood

$$\mathcal{N}_8 = \{(-1,-1), (-1,0), (-1,1), (0,-1), (0,1), (1,-1), (1,0), (1,1)\}. \tag{23}$$

After completing the cumulative cost field search, the refined seamline sequence is obtained by backtracking from the endpoint along the set of predecessor pointers $Q = (u, v) \to (i, j)$

$$\mathbf{S}_{\mathrm{fine}} = \left\{ \left(i_k^*, j_k^*\right) \mid k = 0, \ldots, K^* \right\}. \tag{24}$$

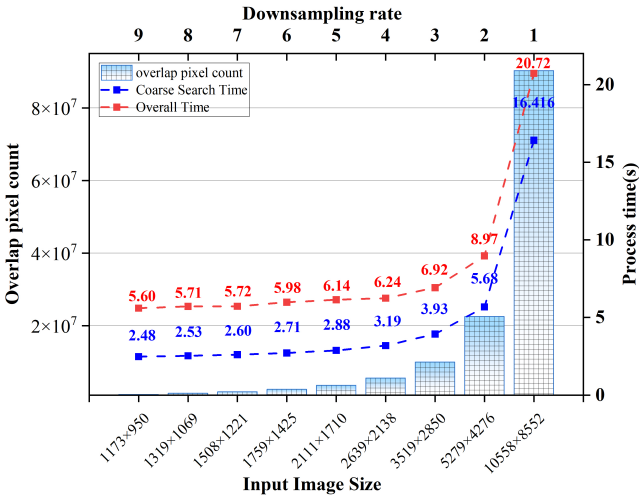

Fig. 8. Relationship between the number of pixels in overlapping regions and the processing time.

D. Downsampling of the Overlapping Image Region (Optional)

To further accelerate the computation of coarse seamline extraction over large-scale remote sensing images, a downsampling operation can be optionally applied to the overlapping regions prior to the first-step detection. For large-area, high-resolution images, directly performing DP on the original pixel level can result in a computational burden that grows exponentially with the image size. By applying a certain level of downsampling to the overlapping areas, the number of pixel nodes is significantly reduced, thereby improving the efficiency of coarse seamline extraction. To validate this, we conducted experiments on a representative urban region with dense buildings, evaluating performance under various downsampling factors.

The horizontal axis indicates different downsampling factors (with the inverse shown in the upper-right corner) [see Fig. 8], along with the corresponding original image sizes. The left vertical axis (blue bars) represents the total

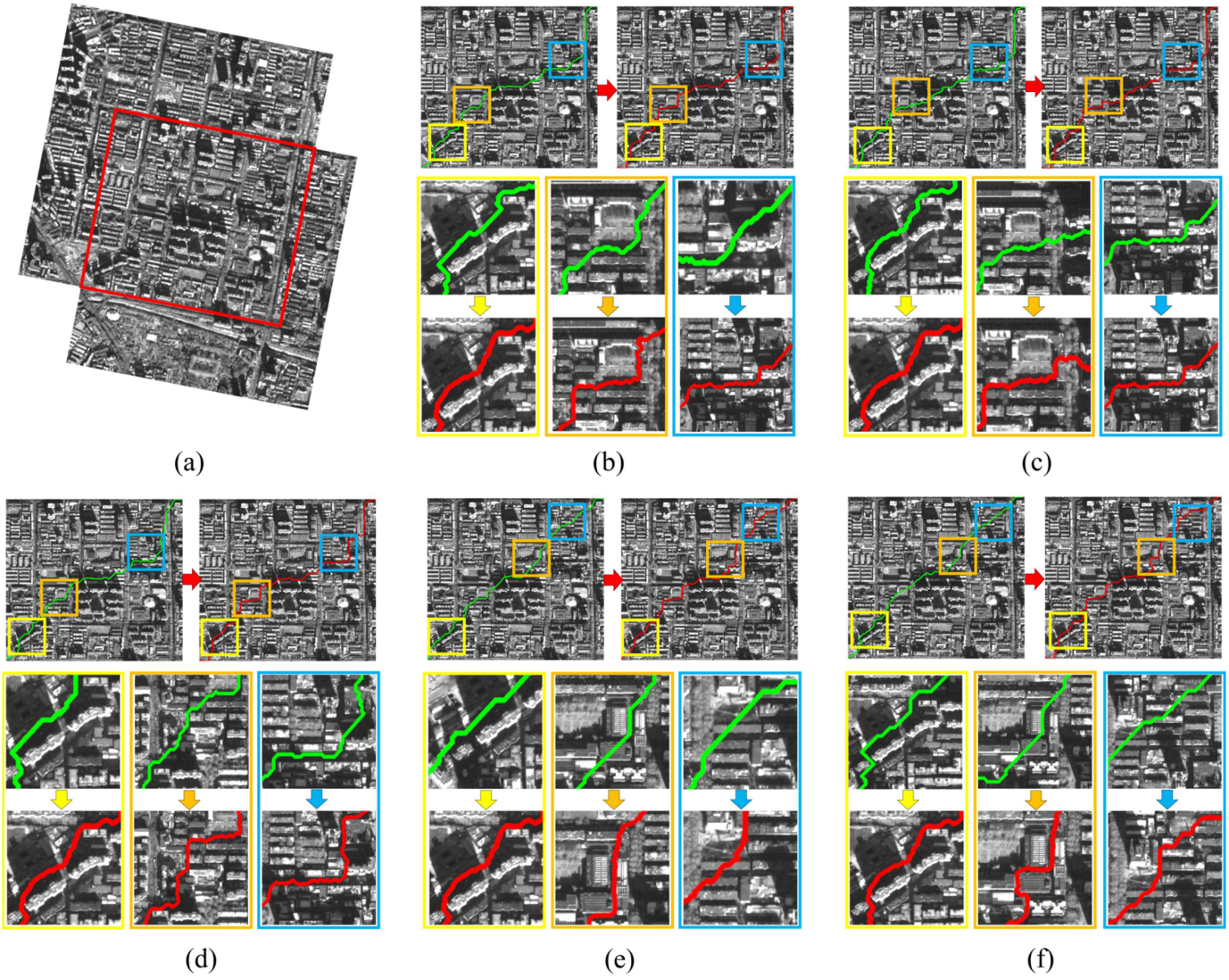

Fig. 9. Seamline detection under varying downsampling factors in a dense urban overlap region. (a) Overlapping region in a densely built urban area. (b) Seamline detection without downsampling. (c) Seamline detection after 3× downsampling. (d) Seamline detection after 5× downsampling. (e) Seamline detection after 7× downsampling. (f) Seamline detection after 9× downsampling.

number of pixels in the overlapping regions after downsampling, while the right vertical axis displays the runtime for "seamline detection" (blue dashed line) and "entire pipeline" (red solid line), respectively. As the input image resolution decreases progressively from 10^8 to 10^6 pixels, the runtime drops exponentially, consistent with the reduction in pixel count. However, since the processing pipeline includes fixed-cost operations such as image loading and fine seamline detection—which are independent of pixel count—these components start to dominate the total runtime as downsampling increases, causing the overall execution time to flatten out.

The results of seamline detection under different resolutions are illustrated for a densely built-up urban area [see Fig. 9]. The number of buildings intersected by the seamline is generally positively correlated with the downsampling factor. When the downsampling factor is 1 (i.e., full resolution), the seamline is searched directly on the original image. Owing to the global optimality of the DP-based method, the first step already yields a satisfactory result, and the second-step refinement further improves both coarse and fine seamline detection, achieving the best overall performance.

For downsampling factors between 1× and 4×, object contours remain relatively clear, and the seamline detection results are still acceptable. However, when the downsampling factor increases further, image details such as object boundaries become significantly blurred, which degrades the effectiveness of coarse seamline detection. As a result, the coarse seamline tends to traverse building clusters. Furthermore, due to the limited width of the buffer zone used in the second-step refinement, the fine optimization fails to avoid these structures. Fig. 10 also demonstrates the positive correlation between the downsampling factor and the number of buildings crossed by the seamline.

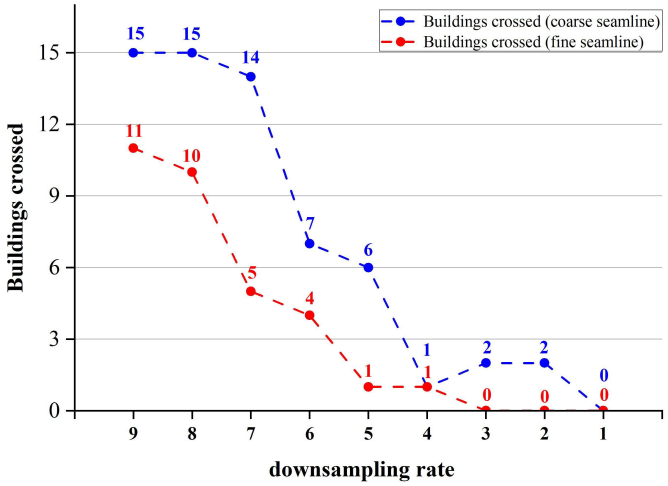

Fig. 10. Relationship between downsampling factor and number of buildings crossed by the seamline.

IV. Results and Discussion

To ensure a fair comparison, we first reimplemented several mainstream mosaicking techniques for optical remote-sensing imagery that are widely used for large-scale production and used them as baseline methods. Subsequently, we conducted experiments on: 1) GF-7 panchromatic imagery, orthorectified and resampled to a 1 m resolution, covering densely built-up urban areas; 2) ZY-3 imagery, orthorectified and resampled to 4 m, covering suburban villages; 3) GF-1 imagery, orthorectified and resampled to 19 m, covering a heterogeneous region with urban, rural, and fluvial features; and 4) GF-3 SAR imagery, which was first geolocated using an optical reference map, tie-point matched and block-adjusted, orthorectified, and resampled to 3 m resolution, to evaluate the applicability and robustness of the proposed method under severe speckle noise and complex texture conditions. For each scenario, locally enlarged views and quantitative metrics were employed to evaluate seamline-detection performance, and three additional regional datasets were analyzed to further illustrate the method's behavior under varied land-cover conditions. To assess scalability and robustness on ultralarge datasets, we subsequently performed a batch mosaicking experiment on 627 GF-7 panchromatic images covering the entirety of Henan Province. All algorithms were developed and executed in C++ under Visual Studio 2017 on a Windows 11 workstation equipped with an Intel i5-13500HX CPU (2.5 GHz) and 32 GB RAM, with the compiler optimization set to /O2 (maximize speed), ensuring reproducibility of the reported results.

A. Comparative Experiments and Analysis

1) Qualitative Evaluation of Geometric Continuity: We reproduced several existing mosaicking techniques suitable for large-scale remote sensing imagery, selecting test regions from GF-7 urban dense areas, ZY-3 suburban village areas, and GF-1 scenes containing a rich mix of urban, village, and river features [see Fig. 11].

The SMP-DP algorithm employs a multidirectional DP approach, using classical cost terms such as edge information extraction, grayscale difference, and gradient difference, and supports fast path search on large 2-D matrices. The Dijkstra algorithm applies the fine-level seamline search method described above. The A* method constructs a heuristic cost function based on grayscale and gradient differences; by maintaining open and closed sets, it greatly reduces the number of search nodes and further accelerates computation via a priority-queue data structure. In the graph-cut–based approach, we first adopt Li et al.'s weighting scheme based on gradient and texture information, then follow Chai et al.'s two-stage framework: an initial seamline search on a downsampled image, followed by a buffered search on the full-resolution image to extract pixel-level seamlines, thereby discarding most "invalid" nodes. We also include a comparison with the commercial software OrthoVista. Quantitative comparisons in terms of runtime and feature-crossing counts are reported in Table I.

Fig. 12 compares the performance of six seamline detection methods in the same GF-1 urban–village mixed scene. The red curves indicate the final seamlines obtained by each algorithm. Yellow boxes 1–3 in the main image highlight local regions; their magnified views are shown sequentially along the right and bottom for detailed comparison. Our method, by virtue of its global optimality, routes the seamline along rivers and open areas and automatically conforms to roads or riverbanks at urban–water and slope–forest transitions. The path is smooth and largely avoids high-texture building clusters and regions of severe color contrast. Compared with our method, SMP-DP's search directions are insufficiently flexible: in regions 1 and 2, it fails to follow the riverbank and instead crosses many features; in the narrow valley of region 3, it even cuts across the riverbank. The Dijkstra method, which exhaustively searches the entire overlap area and is sensitive to edges, produces seamlines that adhere to riverbanks and generally avoid high-texture building clusters and strong color-contrast edges. The nonstandard weighted A* method achieves only approximate optimality: its seamline misses potential best paths, crossing more features in regions 1 and 2 instead of closely following the river, and in region 3, only partially hugging the riverbank while still traversing many features. The graph-cut method suffers from high memory usage and time cost; after downsampling, its ability to recognize texture information is greatly reduced. It fails to follow riverbanks in regions 1 and 2, crosses many features, and in region 3, again, only partially conforms to the riverbank while traversing many features. OrthoVista's commercial software produces seamlines with obvious block-shaped corners and insufficient detail; although it partially follows the riverbank in regions 2 and 3, it does not fully leverage texture information to avoid complex objects. By embedding gradient magnitude and orientation into the cost function, the algorithm naturally steers the seamline away from genuine object boundaries, preventing it from crossing misaligned features. A local-entropy term assigns high costs to texture-rich regions, while the radiometric-difference term drives costs in road and river zones to very low values. The resulting composite cost surface therefore attains its minima along road medians, valley bottoms, and open hillsides, so the search intrinsically traces these linear, low-gradient elements

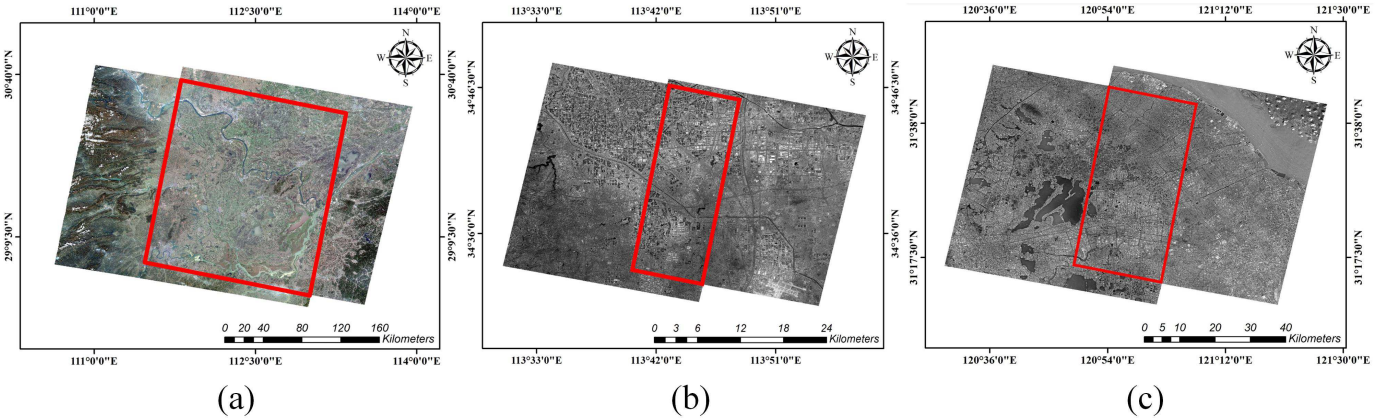

Fig. 11. (a) GF-1 imagery overlapping region. (b) GF-7 imagery overlapping region. (c) ZY-3 imagery overlapping region.

and yields tighter adherence. In urban imagery, the seamline tends to thread through roadway centers; in river networks, it follows the thalweg or mid-channel sandbars, avoiding abrupt edge transitions. Extensive tests over districts, waterways, and sloping terrain confirm the proposed method's broad generality and superior conformity, producing markedly closer fits to road center-lines and riverbanks than existing alternative schemes.

Fig. 13 presents GF-7 imagery over a densely built urban area, shown here with local zoom-ins for detailed comparison. In Fig. 13(a), our proposed method's seamline avoids texture-intensive building clusters and consistently follows the road centerline: in region 1 it detours along the outside of a ring road, avoiding rooftop-dense stands; in region 2 it adheres tightly to the road through a dense residential block without cutting through any buildings; in region 3 it threads through the gap in an interchange where roads intersect; and in region 4 it again follows the road through a sparsely built industrial park. Fig. 13(b) shows the traditional SMP-DP algorithm. In region 2, it crosses through the building cluster, and because its search directions lack flexibility, it fails to follow the road in region 3 and cuts across another building cluster. Fig. 13(c) shows results using our Dijkstra-based refinement. It directly cuts through the building cluster in region 2, and because it is overly sensitive to minor cost fluctuations, it exhibits repeated zig-zagging in regions 3 and 4, resulting in redundant detours. Fig. 13(d) shows the A* method with fixed grayscale–gradient weights. Lacking sensitivity to edge information, it crosses many features in areas where grayscale changes are subtle. Fig. 13(e) shows the graph-cut approach. Although it guarantees global optimality, heavy downsampling impairs its ability to detect edges; while it avoids building clusters to some extent, it still cuts across the overpass in regions 2 and 3. Finally, Fig. 13(f) shows the commercial software OrthoVista. Its seamlines cross prominent buildings in regions 1 and 3, and in region 4—where edge gradients are weak—it likewise traverses major structures. The proposed method consistently delivers the highest positional accuracy and obstacle-avoidance performance across diverse urban features—including high-density residential blocks, expressway interchanges, and large industrial complexes—while keeping the seamline closely aligned with road centerlines. By incorporating an entropy-based cost term, the algorithm is further able to steer the seamline away from dense building clusters whenever possible. In contrast, traditional DP, Dijkstra, A*, unregularized graph-cut approaches, and several commercial software packages all produce seamlines that penetrate buildings or cross major roads in at least one or two test scenes; once a seamline enters a congested built-up area, it inevitably intersects individual structures. These observations underscore the clear advantage of the proposed multistage local-refinement framework when operating in complex urban textures.

Fig. 14 shows a ZY-3 suburban image with sparse buildings, using local zoom-ins for detail. Fig. 14(a) displays the proposed method. In region 1, it follows the road to avoid both building clusters and farmland; in regions 2, 3, and 4, it follows roads and rivers to bypass nearly all buildings. Fig. 14(b) shows the traditional SMP-DP algorithm, which crosses building clusters in regions 2 and 3. Fig. 14(c) shows our Dijkstra-based approach; because it is overly sensitive to minor cost fluctuations, its overall path drifts toward the image boundary and cuts through part of the texture-dense pond in region 2, despite flexibly avoiding major building clusters. Fig. 14(d) presents the A* algorithm, which—lacking edge sensitivity—crosses many features in the low-contrast farmland of region 1 and cuts through some buildings in region 4. Fig. 14(e) shows the graph-cut result; heavy downsampling causes it to cross numerous features in the generally low-grayscale farmland of region 1 and the pond area of region 2. Finally, Fig. 14(f) shows the OrthoVista commercial software result, where the seamline crosses prominent features in regions 1 and 2, fails to follow the river in region 3, and traverses most buildings in region 4. This comparative experiment on a ZY-3 suburban sparse-building area demonstrates that our method consistently follows roads and rivers to avoid obstacles—farmland, channels, and buildings—with high precision in every zoomed region.

To systematically evaluate runtime performance at scale, we constructed a nine-level, progressive downsampling dataset—from 3× down to the original resolution (1×)—over a fixed overlap region, reporting computation time only (excluding disk I/O) [see Table II].

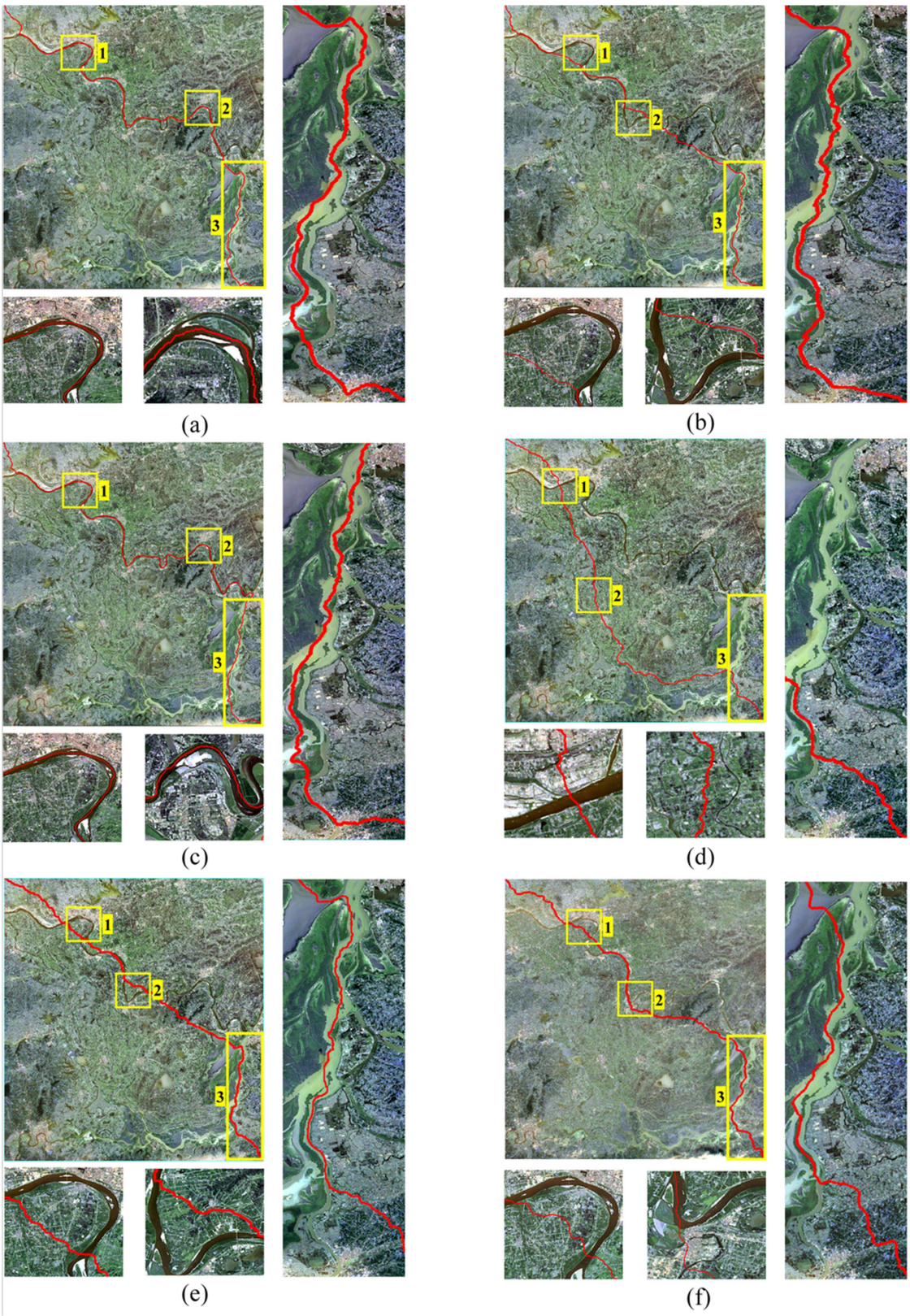

Fig. 12. Seamline detection results for different methods in the GF-1 imagery overlapping region. (a) Proposed method. (b) SMP-DP method. (c) Dijkstra-based method. (d) A*-based method. (e) Graph-cut-based method. (f) OrthoVista.

Experimental results show that execution time increases exponentially with pixel count. As the downsampling factor is reduced from 3× back to 1×, the runtimes of Dijkstra, A*, and the commercial OrthoVista software all exhibit exponential growth, whereas our DP-based method and SMP-DP follow markedly gentler curves [see Fig. 15]. At full

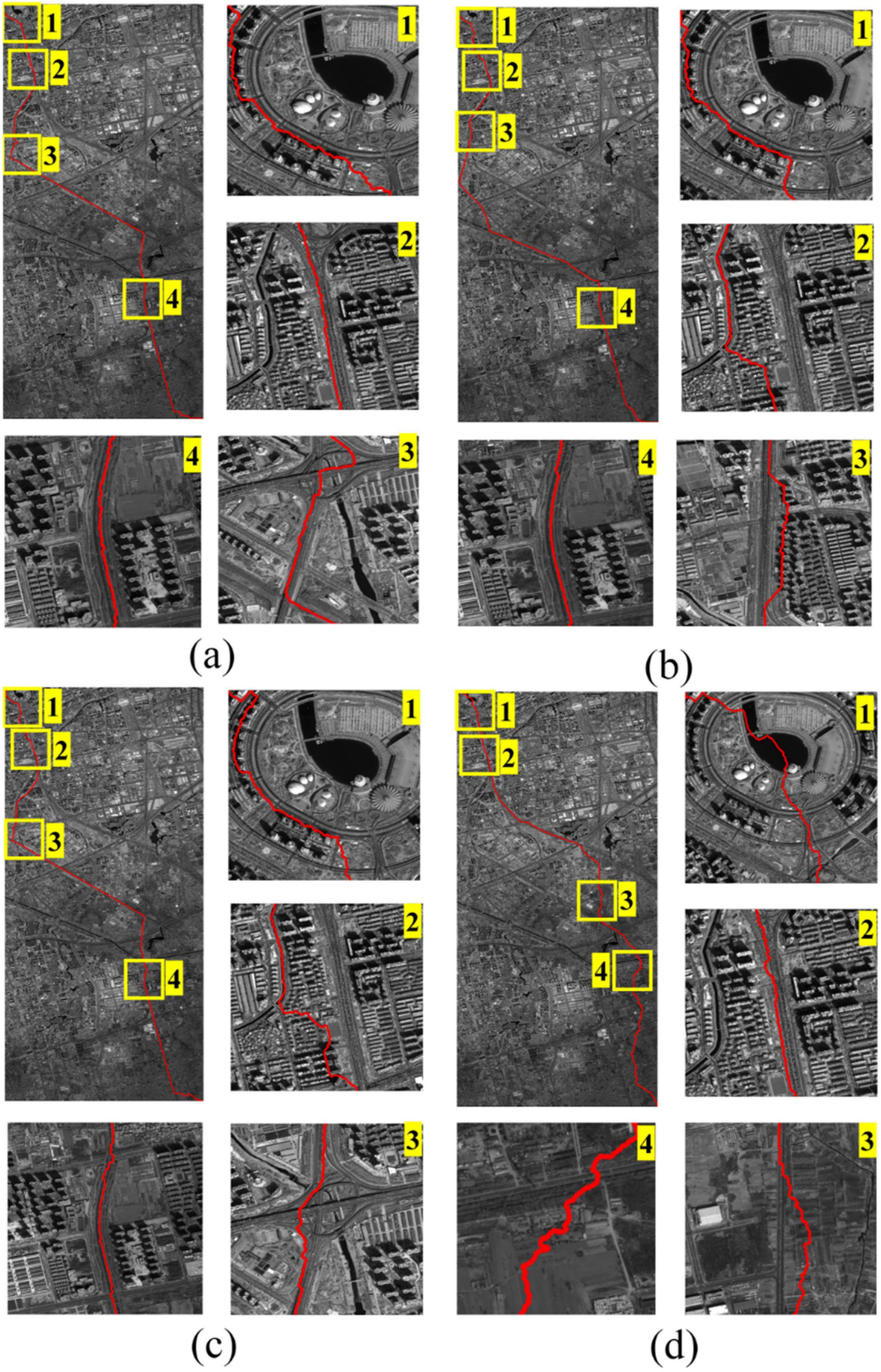

Fig. 13. Seamline detection results for different methods in the GF-7 imagery overlapping region. (a) Proposed method. (b) SMP-DP method. (c) Dijkstra-based method. (d) A*-based method.

resolution (1×), our method completes in 90 s—compared to 112 s for SMP-DP, 259 s for Dijkstra, 173 s for A*, and 223 s for OrthoVista—yielding speedups of approximately 1.2×, 2.9×, 1.9×, and 2.5×, respectively. At 1.5×

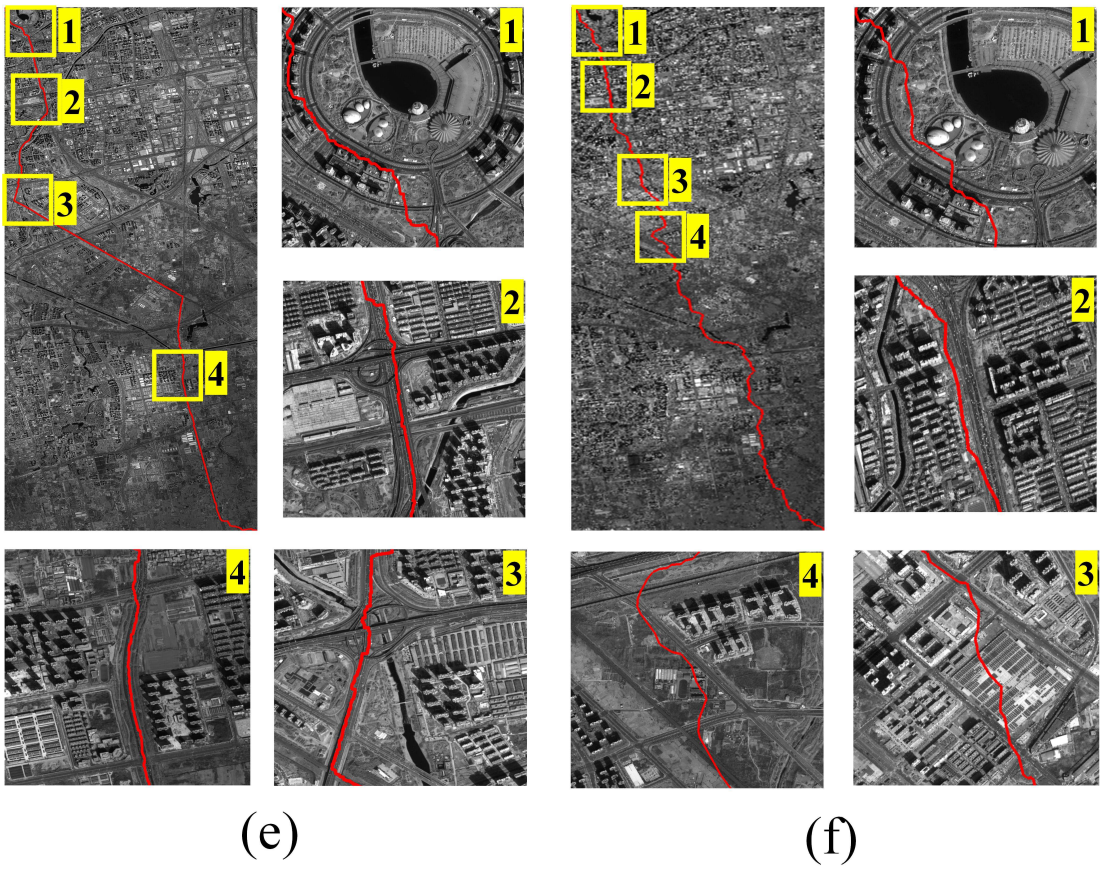

Fig. 13. *(Continued.)*Seamline detection results for different methods in the GF-7 imagery overlapping region. (e) Graph-cut-based method. (f) OrthoVista.

downsampling, our algorithm requires only 27.9 s, still significantly faster than SMP-DP (46.7 s) and OrthoVista (95 s). Under the maximum downsampling (3×), we achieve 6.9 s, whereas OrthoVista takes 22 s, Dijkstra 18.6 s, and A* 10.1 s. Notably, graph-cut–based methods—with worst case complexity $O(\mathrm{NM}\sqrt{NM})$—exceed 500 s across all tested resolutions, making them infeasible for large-scale mosaicking. Given the exponential runtime trends, A* and Dijkstra [both $O(\mathrm{NM}\log(\mathrm{NM}))$] would become unacceptable for still larger swaths. In contrast, our method consistently achieves the lowest runtime at every scale, and its linear complexity $O(\mathrm{NM})$ confers an ever-increasing relative advantage as overlap regions continue to expand.

When the overlap region reaches $N \times M \approx 10^8$ pixels, any algorithm with a superlinear factor (e.g., log(NM) or $(NM)^{1/2}$) quickly amplifies its workload at scale and thus dominates compute time. A* and Dijkstra incur $O(\mathrm{NM}\log(\mathrm{NM}))$ complexity due to priority-queue maintenance, where the log(NM) term is a nonnegligible multiplier; Graph-Cut requires constructing a dense grid graph and solving repeated min-cuts with worst case complexity $O(\mathrm{NM}(NM)^{1/2})$. By contrast, our global cumulative-cost DP is strictly linear, $O(\mathrm{NM})$, performing only one addition and one comparison per pixel; the local refinement stage is confined to a narrow buffer of size $B = O(10^6)$, and its weighted Dijkstra costs $O(B \log B)$, which does not change the overall $O(\mathrm{NM})$ order. Consequently, at a large scale, the inevitable main cost arises from the per-pixel linear scan of the DP core (an algorithmically linear workload), whereas competing methods with superlinear factors exhibit faster growth and larger constant terms in both theory and practice. This complexity-based analysis shows that the efficiency bottleneck is fundamentally determined by algorithmic order and irreducible operation counts.

In conclusion, the proposed two-stage heuristic search framework combines global optimality with highly flexible search directions. In the global coarse-search stage, the improved acyclic six-direction DP guarantees a path of minimum cumulative cost, achieving large-scale global optimality across various features (roads, rivers, farmland, and building clusters). In the local refinement stage, a weighted Dijkstra search based on grayscale difference, gradient magnitude, and edge information performs fine adjustments within the buffer zone, precisely following roads and waterbanks. Compared to traditional SMP-DP, which is limited by fixed traversal directions; A*, which depends on heuristic accuracy and is difficult to tune; pure Dijkstra and Graph-Cut, which incur high computational and memory costs on large images; and closed-source tools like OrthoVista, which lack customizability—our method demonstrates superior path optimality, stronger local obstacle-avoidance capability, and higher runtime efficiency across multiple scenarios, fully validating its significant advantages in large-scale remote sensing mosaicking.

2) Quantitative Evaluation of Geometric Continuity: To comprehensively evaluate the geometric continuity of

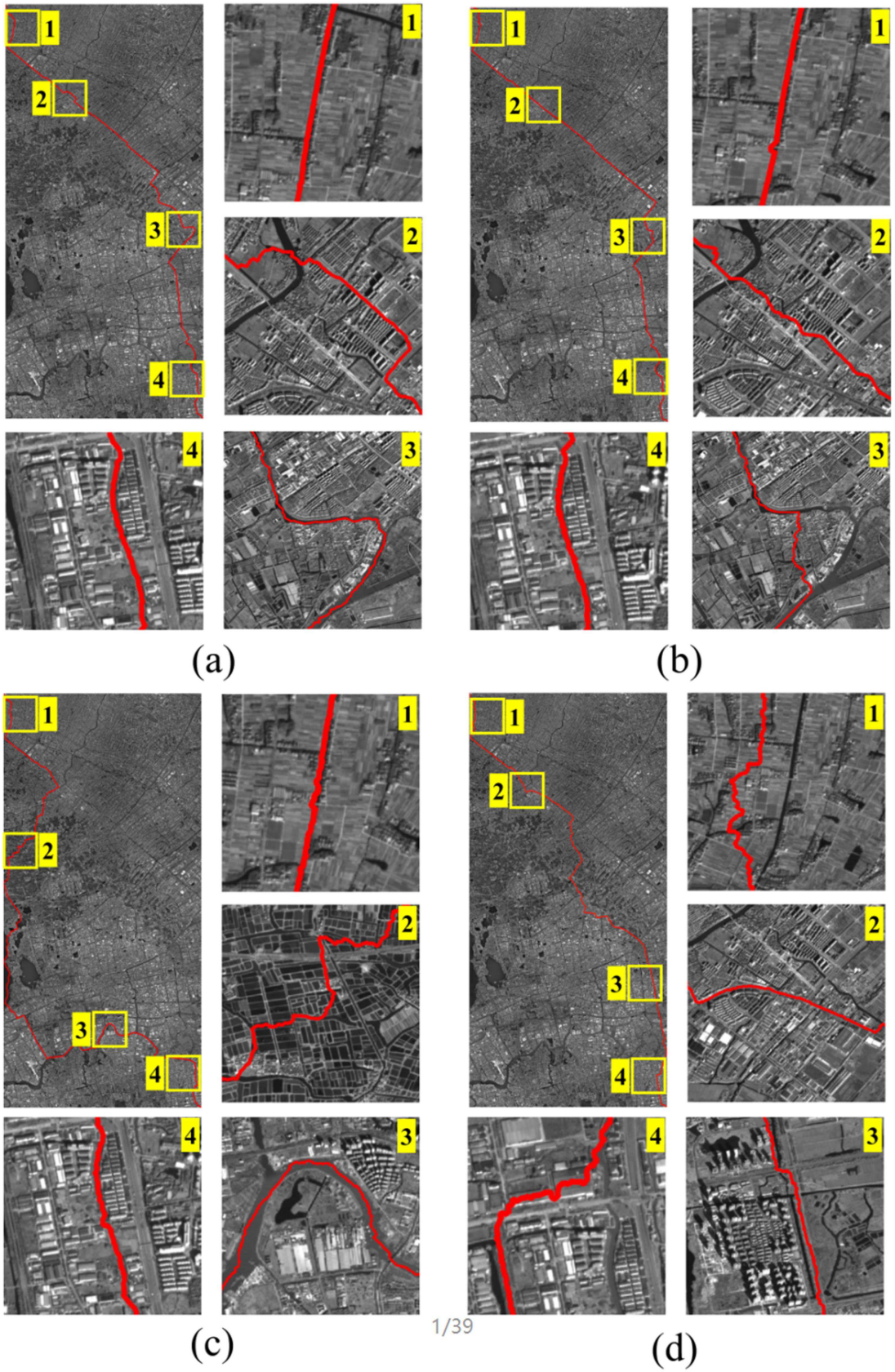

Fig. 14. Seamline detection results for different methods in the ZY-3 imagery overlapping region. (a) Proposed method. (b) SMP-DP method. (c) Dijkstra-based method. (d) A*-based method.

seamlines in large-format remote-sensing mosaics, an indicator must deliver: 1) pixel-level precision; 2) robustness across heterogeneous land-cover textures; and 3) fully automatic, repeatable operation. Conventional practice—manually

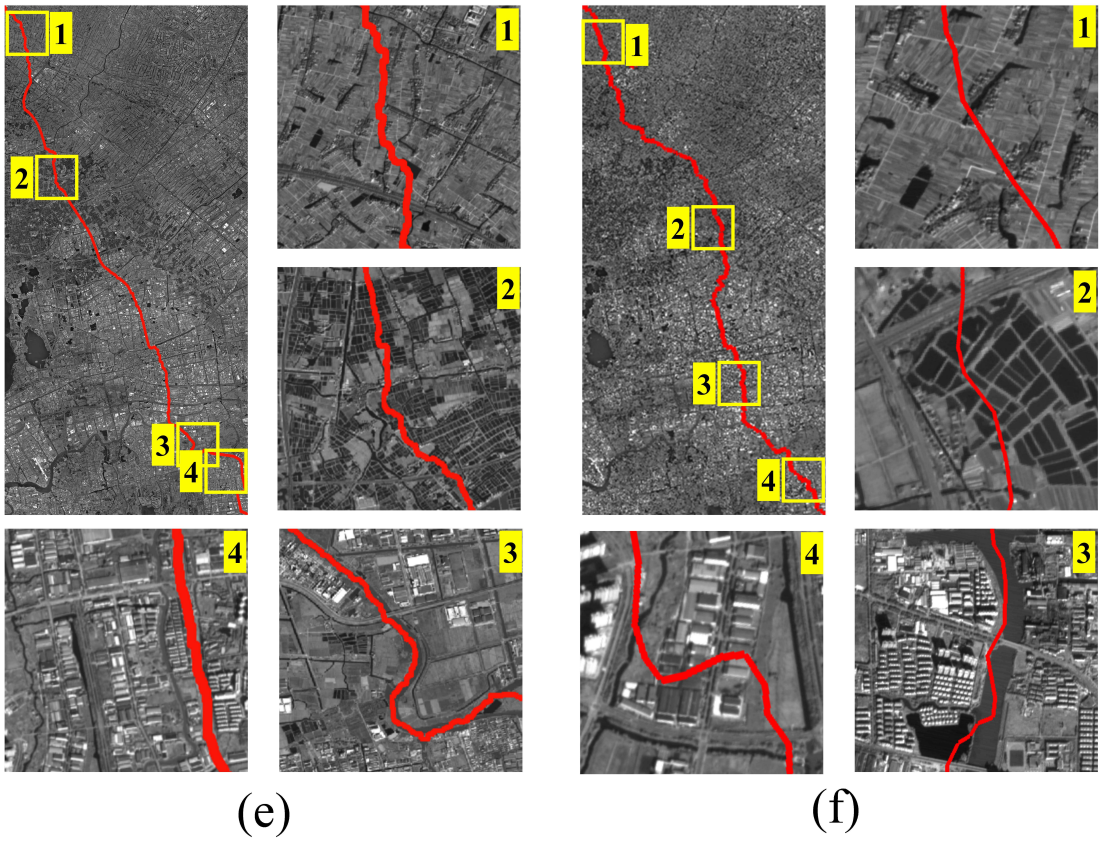

Fig. 14. *(Continued.)* Seamline detection results for different methods in the ZY-3 imagery overlapping region. (e) Graph-cut-based method. (f) OrthoVista.

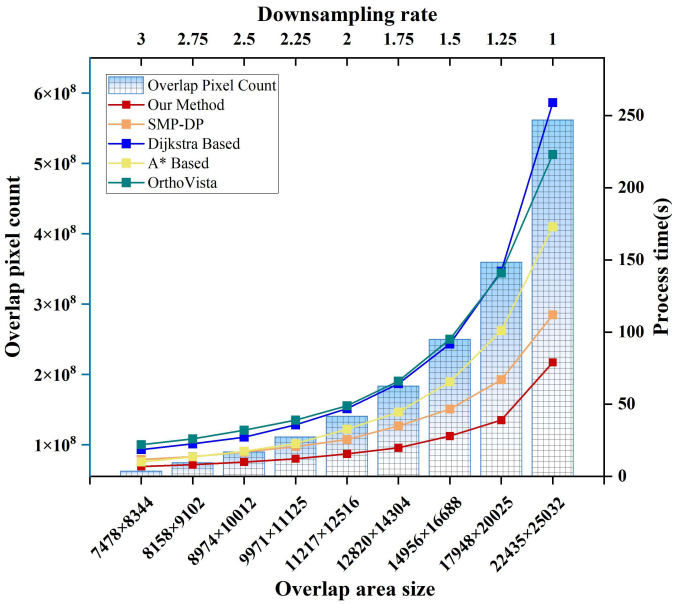

Fig. 15. Execution times of various methods across overlap regions of different sizes.

counting misaligned buildings—is time-consuming and highly susceptible to subjective bias. To overcome these limitations, we adopt the SIFT descriptor as the evaluation core: by performing high-dimensional statistics on the gradient-orientation distribution within each keypoint's neighborhood, the method precisely quantifies the local structure of the seamline. After L2 normalization and threshold truncation, SIFT descriptors remove the effects of brightness and contrast changes, retaining only structural orientation information. Their 128-D histograms are extremely sensitive to subpixel shifts, so even minute geometric displacements accumulate across multiple subregions and are accurately captured through the Euclidean distance between feature vectors. Compared with manual visual tallies of "misaligned building counts," the proposed metric requires no human intervention, provides objective and reproducible assessments, eliminates subjective bias and inefficiency, and is easily scalable to large-scale batch processing (see Fig. 16 for the evaluation flowchart). Let the set of seamline pixels be

$$\mathcal{S} = \{p_i \mid i = 1, 2, \ldots, N\}. \tag{25}$$

First, at the same point p_i in both images I_A and I_B, compute the Sobel gradient magnitude denoted by $G(p_i)$, and set a threshold $T = 10$: when $G(p_i) < T$, it is treated as a flat region, and the 128-D SIFT descriptors at that point in both images are directly set to zero vectors, $\mathbf{d}i^A = \mathbf{0}, \mathbf{d}i^B = \mathbf{0}$, eliminating errors caused by the protective constant in the normalization denominator in textureless areas; otherwise, descriptors are extracted according to the standard SIFT computation pipeline.

Within the $k \times k$ window centered at p_i and aligned by scale and main orientation, extract the raw histogram vector

TABLE I
COMPARISON OF RUNTIME AND NUMBER OF CROSSED FEATURES

Dataset	Our Method		SMP-DP [26]		A* Based [16]		Dijkstra Based		Graph-cut Based [21], [22]		OrthoVista	
	Time (s)	Features	Time (s)	Features	Time (s)	Features	Time (s)	Features	Time (s)	Features	Time (s)	Features
GF-1 (9568×10409)	17.5	14	22.5	41	20.5	105	38.0	19	>500	78	104.0	60
GF-7 (11534×25438)	45.9	2	52.6	7	56.8	22	91.6	6	>500	12	282.0	27
ZY-3 (6460×12081)	11.8	2	12.6	19	16.2	16	26.5	27	>500	32	79.0	39

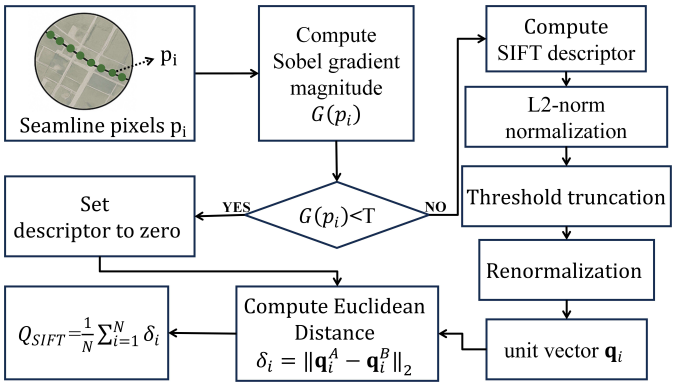

Fig. 16. Flowchart of the geometric continuity evaluation procedure.

$\mathbf{d}i = [di, 1, di, 2, \ldots, di, 128]^T$. k is set to 8. Denote its L2 norm as

$$\|\mathbf{d}_i\|_2 = \sqrt{\sum_{j=1}^{128} d_{i,j}^2}. \tag{26}$$

Subsequently, a two-step normalization is performed

$$\tilde{d}_{i,j} = \frac{d_{i,j}}{\|\mathbf{d}_i\|_2}, \quad j = 1, \ldots, 128 \tag{27}$$

$$\bar{d}_{i,j} = \begin{cases} \tilde{d}_{i,j}, & \tilde{d}_{i,j} \le \tau, \\ \tau, & \tilde{d}_{i,j} > \tau, . \end{cases} \quad \tau = 0.2. \tag{28}$$

In the two-step normalization framework, the descriptor is insensitive to illumination and contrast changes, and the Euclidean distance responds only to geometric and structural differences. The first normalization ensures that the descriptor remains invariant to overall brightness and contrast; the second step truncates all components exceeding the threshold τ to τ. The core purpose of this is: if a single gradient orientation's large component is left unconstrained, it will dominate the overall difference in subsequent distance calculations, thereby masking misalignment information in other orientations; after truncation and renormalization, each orientation component participates in the Euclidean distance with comparable weight in the final descriptor, thus more comprehensively and stably reflecting subtle geometric discontinuities. The resulting final unit-length vector is obtained by

$$\mathbf{q}_i = \frac{\left[\bar{d}_{i,1}, \bar{d}_{i,2}, \ldots, \bar{d}_{i,128}\right]^T}{\|\left[\bar{d}_{i,1}, \bar{d}_{i,2}, \ldots, \bar{d}_{i,128}\right]^T\|_2}. \tag{29}$$

The local misalignment is measured by the Euclidean distance between the final descriptors of the two images at the same point

$$\delta_i = \|\mathbf{q}_i^A - \mathbf{q}_i^B\|_2. \tag{30}$$

TABLE II
RUNTIME OF SEAMLINE DETECTION METHODS UNDER PROGRESSIVE DOWNSAMPLING

Downsampling Rate	Runtime (s)					
	Ours	SMP-DP	Dijkstra	A*	Graph-cut	OrthoVista
1	81.1	112.3	259.8	173.1	>500	223.0
1.25	39.5	67.2	142.2	101.4	>500	141.0
1.50	27.9	46.7	91.8	65.5	>500	95.0
1.75	19.8	35.1	64.2	44.6	>500	66.0
2	15.6	25.6	46.9	32.7	>500	49.0
2.25	12.3	20.7	35.7	22.7	>500	39.0
2.50	10.1	16.9	27.2	17.5	>500	32.0
2.75	8.1	13.7	22.6	13.6	>500	26.0
3	6.9	11.7	18.6	10.1	>500	22.0

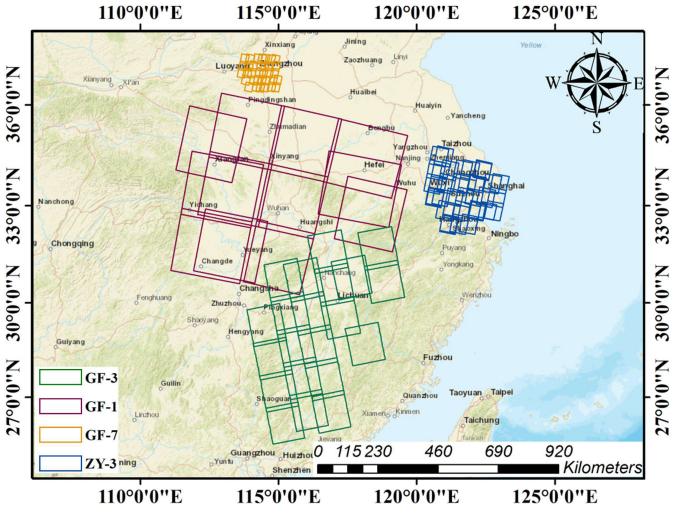

Fig. 17. Schematic of the experimental datasets.

The value of δ_i lies in the interval $[0, 2]$; when perfectly aligned, $\delta_i = 0$; if there is a subpixel-level shift or structural break, the orientation histogram distribution will change significantly, causing δ_i to increase. The overall quality metric is obtained by taking the arithmetic mean of the local distances over all pixels

$$Q_{\text{SIFT}} = \frac{1}{N}\sum_{i=1}^{N}\delta_i. \tag{31}$$

When $Q_{\text{SIFT}} \to 0$, it indicates that the seamline region is highly continuous geometrically; if Q_{SIFT} is significantly larger, it reveals multiple structural misalignments or texture discontinuities along the seamline. Our method combines contrast invariance with sensitivity to subpixel geometric shifts, enabling a stable and accurate assessment of seamline quality in high-resolution remote sensing imagery.

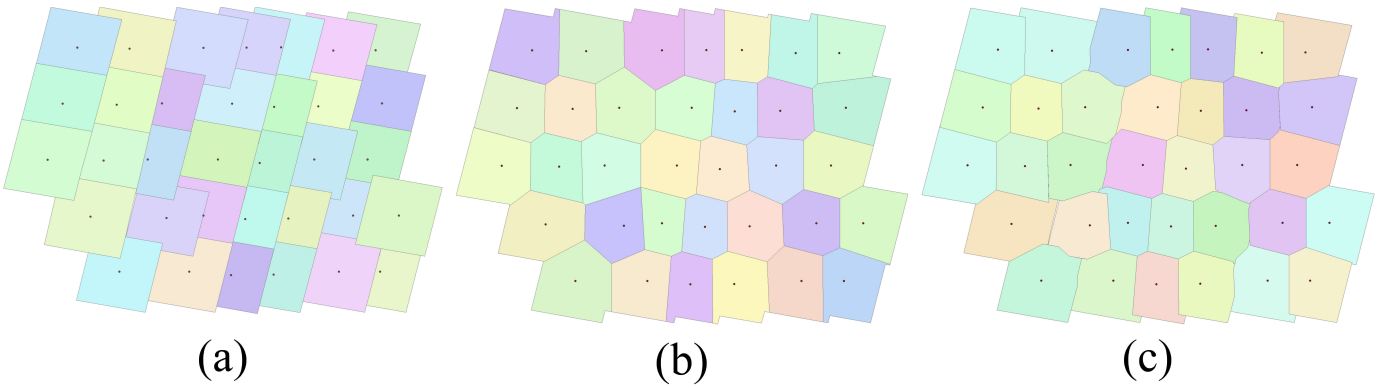

Fig. 18. Initial seamline detection results for GF-7 imagery over Henan region. (a) VRPs of the images. (b) Voronoi diagram. (c) Initial seamline network.

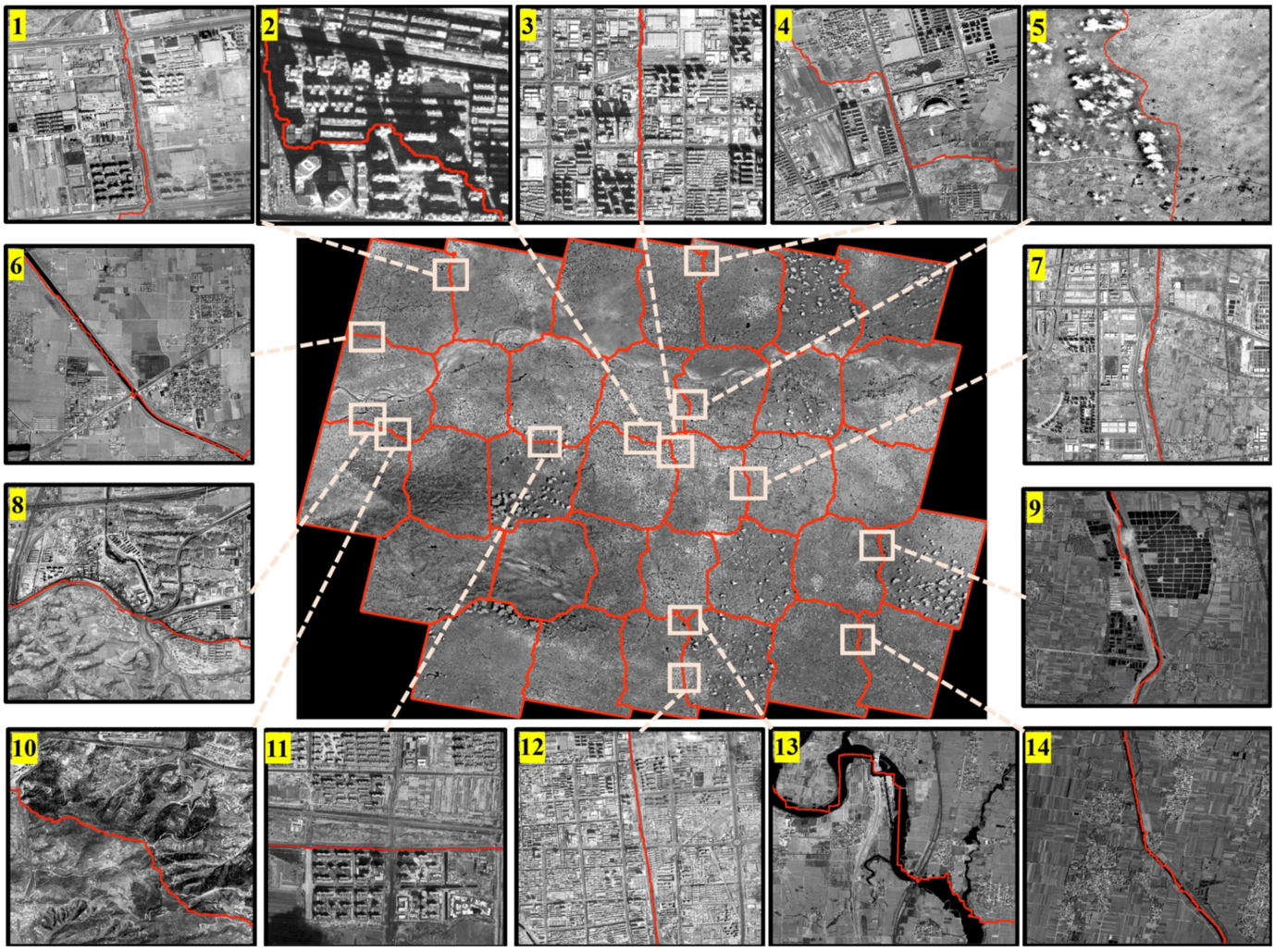

Fig. 19. Seamline detection results for the Henan region using GF-7 imagery.

Table III summarizes Q_{SIFT} scores obtained on three representative remote-sensing datasets—GF-7 (dense urban), ZY-3 (low-density suburban) and GF-1 (village–river composite)—for six competing seamline-detection algorithms.

Across all datasets, the proposed method achieves the lowest Q_{SIFT} (mean = 0.117), clearly outperforming the second-best approach, SMP-DP (mean = 0.201). On the most challenging GF-7 scene the score is reduced by 14.5 % relative to SMP-DP, by 58.7 % and 50.2 % with respect to the single-stage graph-search baselines A* (0.313 276) and Dijkstra (0.259 866), and by 17.8 % and 69.3 % when compared with the globally optimized Graph-Cut (0.157 412) and the commercial package OrthoVista (0.421 519), respectively—demonstrating the ability to maintain continuity while threading accurately

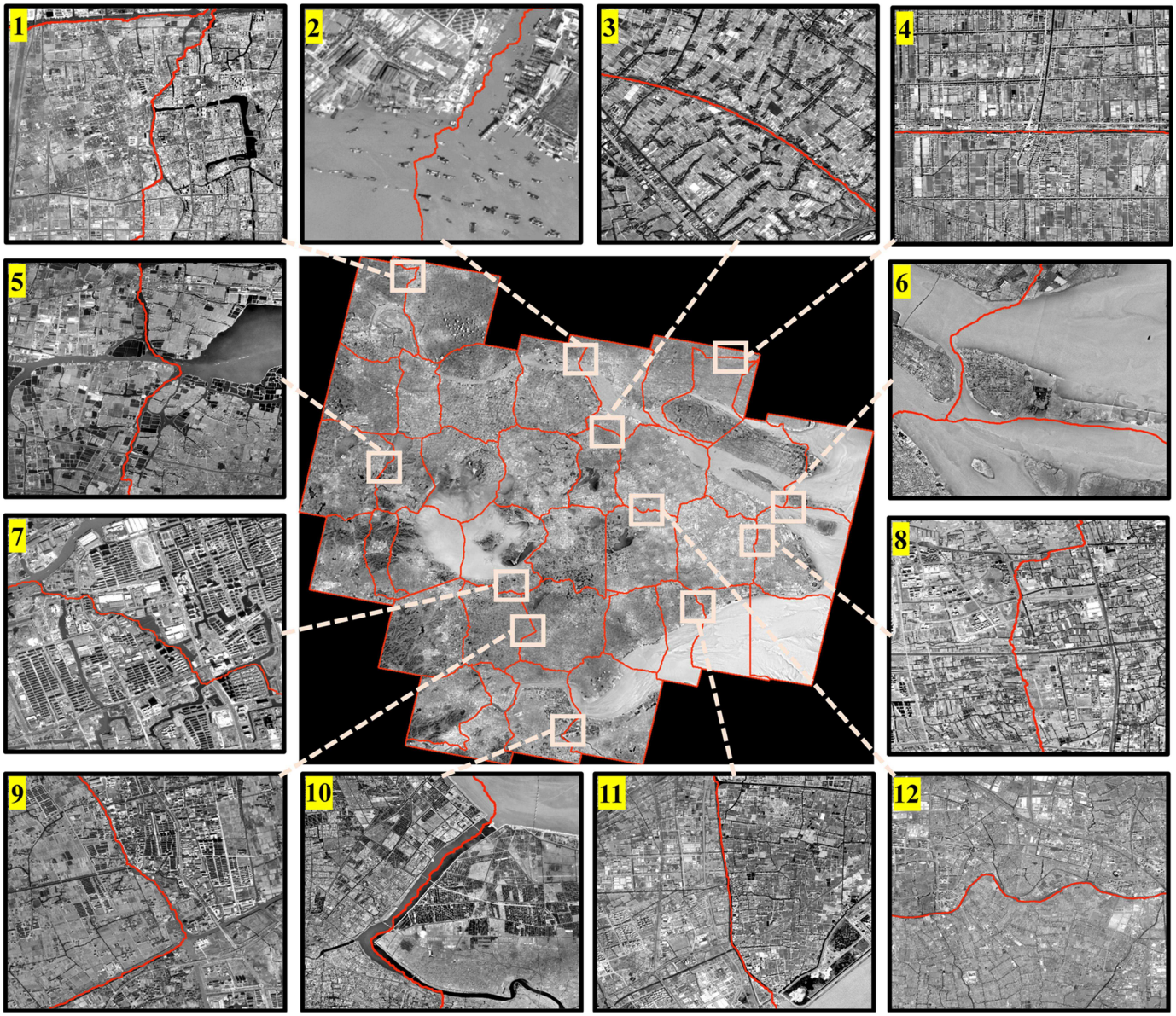

Fig. 20. Seamline detection results for the ZY-3 imagery over the Taihu Lake region.

TABLE III
QUANTITATIVE EVALUATION OF GEOMETRIC CONTINUITY FOR DIFFERENT METHODS

Dataset	Our Method	SMP-DP	A* Based	Dijkstra Based	Graph-cut Based	OrthoVista
GF-7	0.129401	0.151269	0.313276	0.259866	0.157412	0.421519
ZY-3	0.117751	0.120483	0.370498	0.159772	0.394213	0.414798
GF-1	0.105762	0.132493	0.412425	0.108676	0.427854	0.315874

through road centrelines and interbuilding gaps in high-rise districts. In the ZY-3 suburban scene, the proposed score (0.117 751) remains 2.3 % lower than SMP-DP and more than 68 % lower than A* and Graph-Cut; in the GF-1 village–river scene, the margin widens to 20.2 % over SMP-DP and 74.4 % over A*. The interscene variation of the proposed Q_{SIFT} is only 0.023, whereas A* and Graph-Cut fluctuate by more than 0.10, evidencing superior robustness to changes in land-cover texture. This stability arises from: 1) an entropy-weighted cost term that assigns high penalties to complex terrain, such as ridges and cliff faces, steering the seamline away from high-gradient regions and 2) an eight-neighbor Dijkstra refinement confined to a narrow low-gradient buffer that produces pixel-level adherence to road centrelines and river thalwegs.

In summary, the two-step framework—global five-direction DP skeleton followed by narrow-band eight-neighbor Dijkstra refinement, augmented with entropy and gradient-orientation

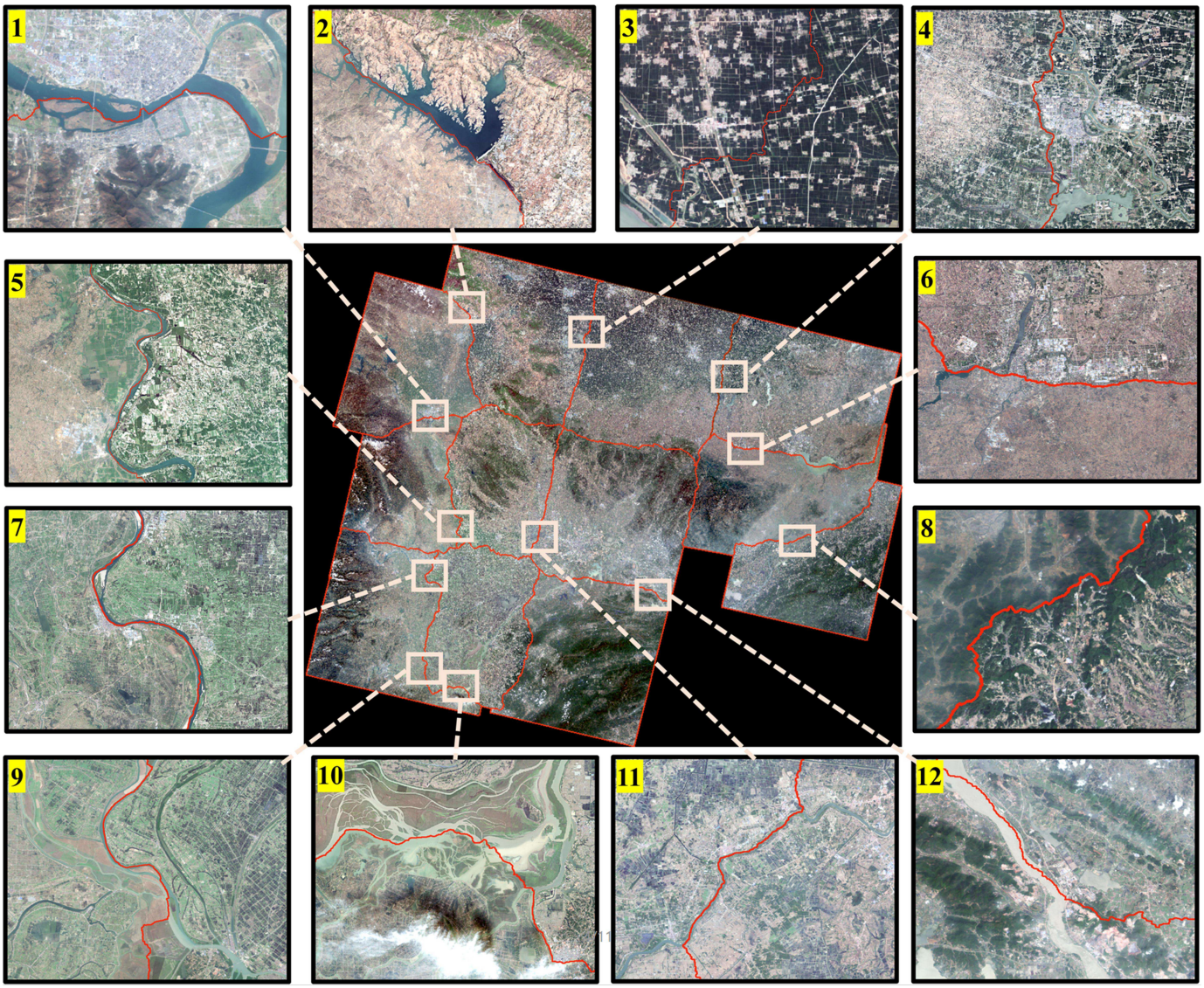

Fig. 21. Seamline detection results for GF-1 imagery over the middle reaches of the Yangtze River.

cues—achieves an optimal balance of accuracy, robustness, and computational efficiency. It significantly surpasses both state-of-the-art academic algorithms and the commercial OrthoVista software, providing a reliable guarantee of geometric continuity for province-scale and larger remote-sensing mosaics.

B. Qualitative Evaluation of Visual Results

Table IV summarizes the dataset counts, mean image sizes, resolutions, sensor types, and runtimes. Fig. 17 presents the experimental results on the GF-3, GF-7, GF-1, and ZY-3 datasets. Fig. 18 illustrates the generation process of the initial seamlines on the GF-7 dataset. In the results obtained on the GF-7 dataset [see Fig. 19], 14 representative regions were selected to demonstrate the effectiveness of the proposed seamline detection method. In densely built urban areas (regions 1, 3, 4, 7, and 12), the proposed method successfully avoids major building clusters and texture-intensive zones by guiding the seamline along roads and open areas, thereby minimizing building misalignment. In region 2, where buildings are extremely dense, the fine-grained Dijkstra-based search within the buffer zone enables the seamline to bypass structures at the pixel level. In region 5, where a large area is covered by clouds, the edge and gradient terms show high sensitivity to cloud regions: the seamline automatically bypasses the cloud clusters, thereby reducing their impact on the final mosaic result. In regions 6, 9, 11, 13, and 14, the seamline traverses along roads and rivers, effectively avoiding densely packed village areas. In regions 8 and 10, located in mountainous terrain, the seamline circumvents texture-heavy ridgelines and instead follows paths along roads and valleys.

In the results obtained on the ZY-3 dataset [see Fig. 20], 12 representative regions were selected to demonstrate the performance of the proposed seamline detection method. In densely built urban areas (regions 1, 4, 7, 8, and 11), the method leverages a globally optimal algorithm to guide the seamline along roads within the city, effectively avoid-

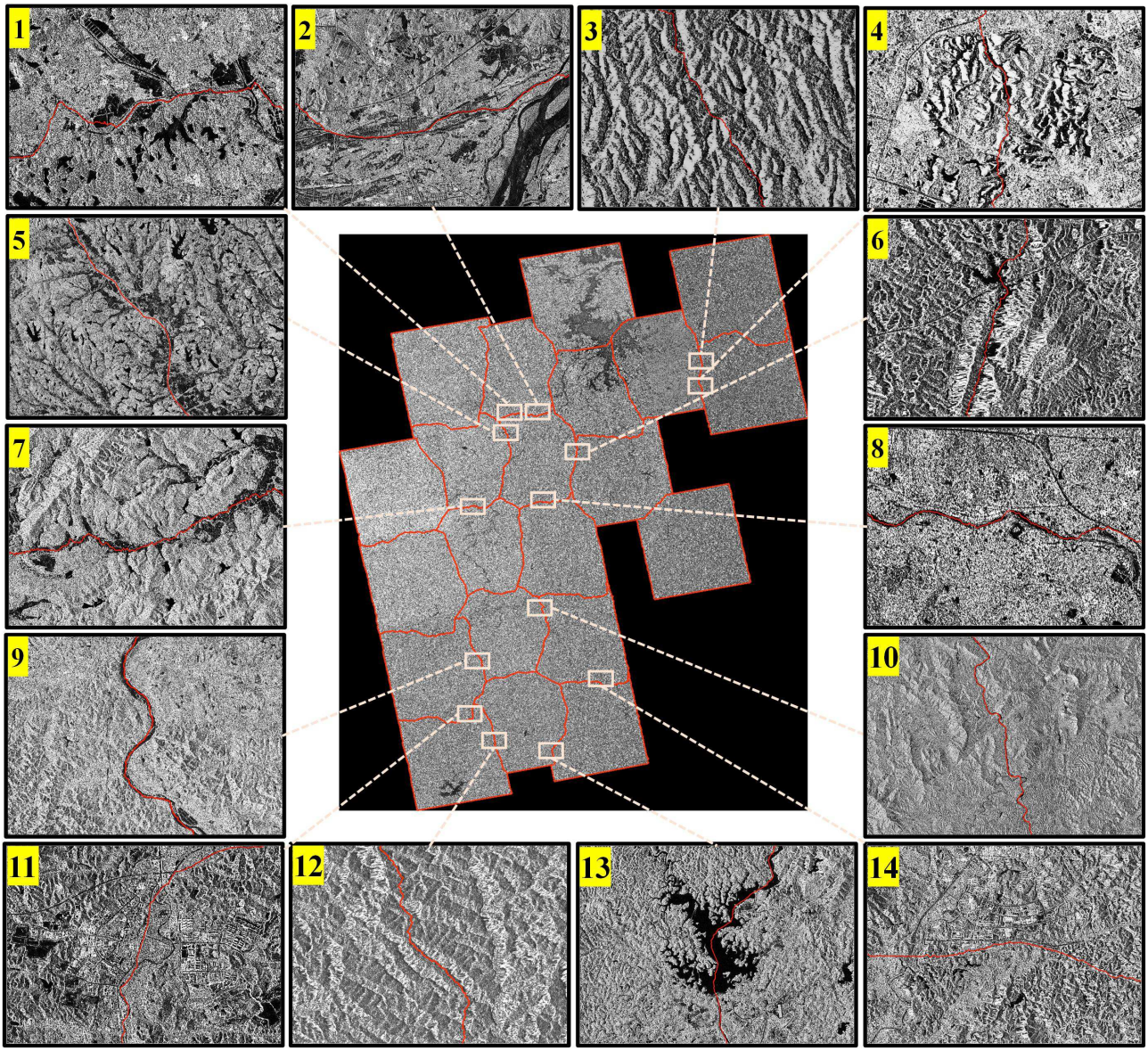

Fig. 22. Seamline detection results for GF-3 imagery over a region in Jiangxi Province.

TABLE IV
PROCESSING STATISTICS OF THREE DATASETS

Dataset	Number	Mean width	Mean height	Orthorectified Resolution (m)	Type	Time (s)
GF-7	34	32082	32545	1	Optical-PAN	826.7
ZY-3	36	17136	16574	4	Optical-PAN	215.2
GF-1	12	13316	13408	19	Optical-MS	72.3
GF-3	21	51255	49747	3	SAR(C-band)	740.2

ing texture-dense building clusters. In rural village regions (regions 3, 5, 9, 10, and 12), the seamline is able to follow roads and rivers, minimizing its passage through farmland and residential areas. In water regions (regions 2 and 6), the seamline adapts to the shape of the waterbody, avoiding islands and ships.

In the results obtained on the GF-1 dataset [see Fig. 21], 12 representative regions were selected to demonstrate the effectiveness of the proposed seamline detection method. In urban areas (regions 1, 4, 5, and 11), the seamlines closely follow roads and rivers, effectively avoiding building clusters with high texture complexity. In mountainous, forested, and rural areas (regions 2, 3, 6, 7, 8, 9, 10, and 12), the seamlines are able to bypass textured structures and traverse through rivers and sparsely textured open areas.

In the results obtained on the GF-3 dataset [see Fig. 22], 14 representative regions were selected to verify the effectiveness of the proposed seamline detection method. In mountainous

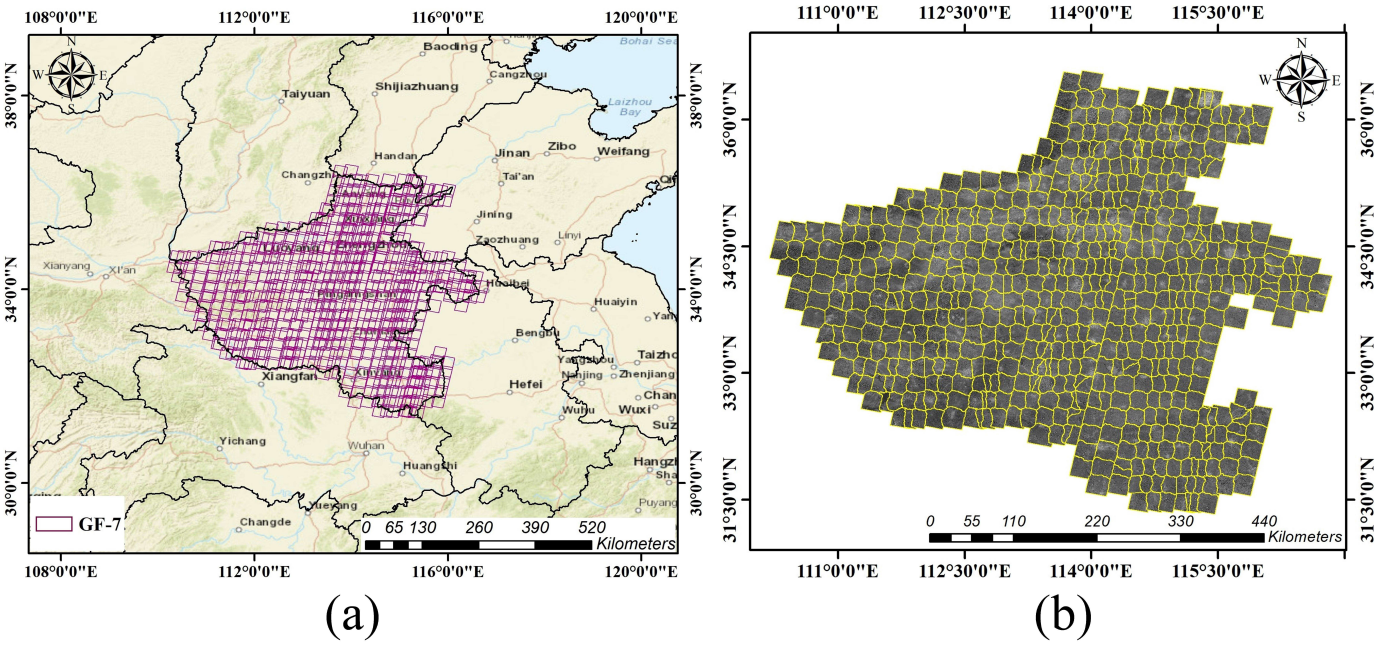

Fig. 23. (a) Administrative divisions of Henan Province, China, and the GF-7 image coverage extent. (b) Resulting seamlines.

TABLE V
STATISTICS OF THE GF-7 DATASET USED IN THE EXPERIMENT

Dataset	Number	Mean width	Mean height	Orthorectified Resolution (m)	Type
GF-7	627	32047	32539	1	PAN

regions (regions 3, 6, and 12), the seamlines naturally follow valleys, avoiding the crossing of prominent ridgelines and thereby minimizing geometric misalignment and visual discontinuities. In the urban region (region 8), the seamline follows the road network. In mixed mountainous–village regions (regions 4, 5, 7, 9, 10, 11, 13, and 14), the seamlines avoid ridgelines and traverse along roads and valleys. In village regions (regions 1 and 2), the seamlines follow roads and avoid passing through the interior of villages, thus preserving the integrity of building shapes and road connectivity.

In summary, across all datasets—GF-1, ZY-3, GF-7, and GF-3—the proposed algorithm consistently aligns seamlines along major linear features such as roads, rivers, and water boundaries across diverse land cover types, including urban areas, villages, mountainous forests, and water bodies. This demonstrates strong adaptability across sensors and scenes. In densely built urban environments, the globally optimal search strategy enables the algorithm to accurately trace road networks while avoiding large-scale, high-frequency textured building clusters. This ensures the seamline remains continuous and semantically reasonable at the macro level, effectively reducing global stitching errors. For extremely dense urban blocks or ridge-dominated mountain regions, the algorithm constructs a local search area using a buffer zone and applies pixel-level Dijkstra refinement within it. This enables the seamline to "adhere" to the edges of roads or rivers, avoiding obstacles with fine-grained precision and achieving accurate pixel-level obstacle avoidance. In relatively sparse-textured areas such as rural villages, farmland, and open terrain, the algorithm automatically selects the minimum-cost path, preferring to pass through open spaces or water edges. This minimizes intersections with houses and agricultural structures, ensuring seamless transitions and high visual consistency. These results demonstrate that the proposed method can generate high-quality seamlines under various terrain conditions, spatial resolutions, and image sources. It significantly reduces manual editing effort and enhances overall production efficiency.

To validate the feasibility and robustness of the proposed seamline detection method for large-scale batch image processing, this study first selected a remote sensing dataset covering the entire Henan Province. Scene counts, sizes, resolutions, and modalities are summarized in Table V. A two-step seamline detection strategy was applied to automatically process all overlapping image pairs across the province in a batch mode. The dataset consisted of GF-7 panchromatic orthophotos with a spatial resolution of 1m, where multiple images overlap arbitrarily in space, collectively covering the full extent of Henan Province [see Fig. 23].

Fig. 24(a) illustrates the Voronoi diagram generated from all image footprints; Fig. 24(b) shows the initial seamline network derived from the image coverage; Fig. 24(c) presents the optimized seamline network based on the initial one; and Fig. 24(d) depicts the final overlay result of the seamlines across the entire province.To fully validate the feasibility and robustness of the proposed two-step seamline-detection framework for ultralarge-scale batch processing of remote-sensing imagery, the study expanded its scope from local scenes to the entire Henan Province. A total of 627 GF-7 Level-3

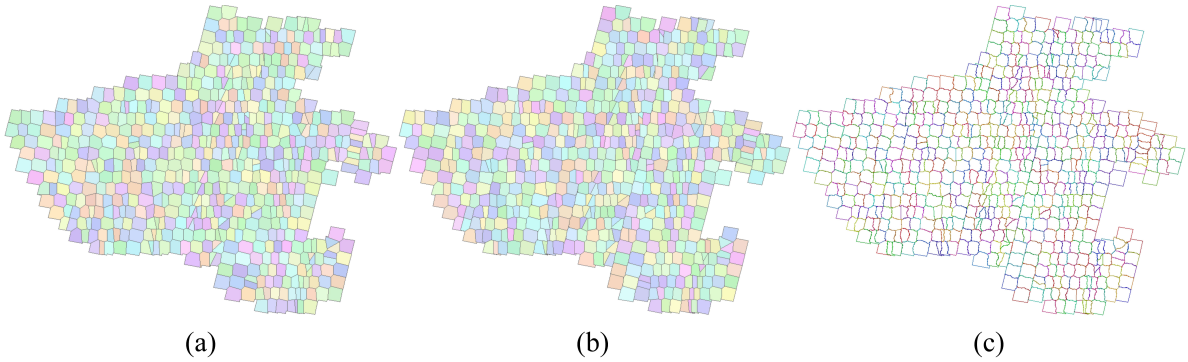

Fig. 24. Seamline detection results for GF-7 imagery covering Henan Province. (a) Voronoi diagram of the overlapping images. (b) Initial seamline network. (c) Optimized (refined) seamline network.

orthophotos with 1 m ground resolution—acquired across multiple seasons and exhibiting arbitrary overlaps—were selected, covering approximately 1.67×10^5 km^2 and containing several trillion pixels, well beyond the capacity of a single machine's memory. The batch workflow runs fully automatically with a single configuration step. First, the valid footprints of all 627 images are reprojected to the WGS-84 geographic coordinate system, and their union is used to generate a global Voronoi diagram [see Fig. 24(a)]. This exclusive-region partition prunes redundant overlap areas, reducing the subsequent search space by 41% relative to naïve pixel-wise processing. Next, initial seamlines are extracted along Voronoi edges for every adjacent-image pair [see Fig. 24(b)]. A three-times down-sampled coarse search followed by a refinement at full resolution is executed in a 14-thread OpenMP environment, yielding a topologically consistent provincial seamline network [see Fig. 24(c)]. On a standard desktop equipped with an Intel i5-13500HX CPU (2.5 GHz) and 32 GB RAM, the peak memory footprint stays below 28 GB, and the complete workflow finishes in 10.6 h, producing 2508 optimized seamline segments with an aggregate length of approximately 8.7×10^4 km. Visual inspection of the overlaid seamlines confirms the absence of self-intersections, gaps, or crossings of prominent object boundaries. To the best of our knowledge, this is the first demonstration of fully automatic, pixel-level seamline extraction over an entire province at 1 m resolution on a single desktop computer, decisively establishing the proposed method's efficiency, scalability, and robustness for real-world production.

V. Conclusion

This article proposed an innovative two-stage, globally optimal seam-line detection framework whose main contributions are as follows: 1) a Voronoi-based iterative cropping scheme drastically reduces data volume, while a coarse search driven by five-direction DP rapidly finds seam-lines that skirt high-texture regions; 2) a pixel-level refinement stage builds a buffer around the coarse path and applies an enhanced Dijkstra algorithm to avoid complex objects precisely, achieving an excellent balance between efficiency and accuracy; 3) a local entropy–gradient composite cost function is introduced to heighten sensitivity to texture-rich and structurally salient areas; and 4) the work presents, for the first time, an objective geometric-continuity metric based on the Euclidean distance between pixel-level SIFT descriptors, eliminating the subjectivity and instability of manual evaluation.

Extensive experiments on three representative datasets—GF-7 (high-resolution), ZY-3 (medium-resolution), and GF-1 (low-resolution)—validate the superiority of the proposed method over five mainstream techniques (SMP-DP, A*, Dijkstra, Graph-Cut, and OrthoVista). The proposed approach improves the geometric-continuity metric by 14.46%, 58.69%, 50.20%, 17.79%, and 69.30%, respectively, while boosting processing efficiency by 12.74%, 19.19%, 49.89%, > 500%, and 83.72%. On the GF-7 urban dataset, continuity gains reach 14.46%, 58.69%, 50.20%, 17.79%, and 69.30%, with corresponding efficiency gains of 12.74%, 19.19%, 49.89%, 90.82%, and 83.72%. For the ZY-3 rural dataset, the improvements in continuity are 2.27%, 68.22%, 26.30%, 70.13%, and 71.61%, and in efficiency 6.35%, 27.16%, 55.47%, 97.64%, and 85.06%. On the GF-1 mixed-terrain dataset, continuity gains are 20.18%, 74.36%, 2.68%, 75.28%, and 66.52%, with efficiency gains of 22.22%, 14.63%, 53.95%, 96.50%, and 83.17%. In all scenarios, the proposed method yields markedly fewer object crossings than its competitors.

Validation on heterogeneous ZY-3, GF-1, and GF-3 imagery, as well as a province-wide GF-7 mosaic, further demonstrates the method's stability and adaptability across resolutions. In dense urban areas and complex textured terrains, the seam-line reliably follows natural texture boundaries, avoiding structural discontinuities and exhibiting outstanding cross-scene robustness. Crucially, the framework operates without DSMs, vector data, or deep-learning semantic priors, and can efficiently support mega-pixel-scale orthophotomosaicking on a single workstation, underscoring its strong practical value.

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